

Distortion of cortical hue geometry in color vision deficiency enables an individualized
color correction filter

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Abstract

Color vision deficiency (CVD) is typically corrected with generic filters designed from a retina-based model. Such filters change how colors appear but improve color discrimination only in some products and tasks. Filter design omits visual cortex, where the retinal loss and the observer’s partial compensation combine. How that combination reshapes color representation in CVD remains underinvestigated. Here we show that CVD distorts the geometry of the cortical color representation rather than weakening its overall signal. We scanned two adults with CVD, one with each red–green subtype, and seven healthy controls. All eight colors remained decodable from cortical activity in both CVD participants, whereas the continuous hue geometry departed from controls, differently in each individual. We modeled each distortion as a hue rotation along two color-space axes and inverted the fit into a personalized stimulus-space filter. Both CVD participants completed a second session of fMRI and behavioral testing to evaluate the filters. Under the individualized filter, the four hue-discrimination thresholds that had exceeded the control range in the first session (three for deutan, one marginal for protan) moved within that range. This small sample cannot establish superiority over a deployed accessibility filter. The direction of cortical change differed across participants and measures, and neither reached the healthy reference. This study identifies a previously uncharacterized geometric distortion of the cortical color representation in CVD and introduces, to our knowledge, the first cortically grounded framework for individualized color correction. Systematic studies can quantify these distortions and provide a new class of personalized filters.

Keywords: color vision deficiency, anomalous trichromacy, fMRI, cortical color representation, individualized color correction

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1 Introduction

Color vision deficiency (CVD) affects approximately 8% of men and 0.4% of women of European ancestry (Birch, 2012). It arises from a shift in the spectral sensitivity of the L- or M-cone photopigment. The magnitude of that shift varies continuously across affected individuals (Bosten, 2019). X-linked polymorphisms in the cone opsin genes underlie the shift (Deeb, 2005; Neitz & Neitz, 2011). The two pigments lie between 1 and 12 nm apart in anomalous trichromacy, compared with roughly 27 nm in normal trichromacy (Somers, Franklin & Bosten, 2024). This narrowed separation attenuates the L–M cone-opponent signal without abolishing it. Red–green CVD comprises two subtypes that differ in the direction of the shift. In protan observers the L-cone sensitivity moves toward the M cone, and in deutan observers the M-cone sensitivity moves toward the L cone (Neitz & Neitz, 2011).

Observers who share a diagnostic label differ substantially in how much their color space is compressed. Anomalous trichromats show reduced red–green discrimination, frequent category confusions, and elevated just-noticeable differences along the L–M chromatic axis. The severity of these signs extends from near-normal to near-dichromatic within one diagnostic subtype (Bosten, 2019; Neitz & Neitz, 2011).

Current CVD correction is calibrated to a population-average retina and applies the same transform to every user. Hardware filters are notch lenses. Each notch attenuates a narrow band of the spectrum where the L and M cones overlap. Removing that band widens the difference between the two cone signals (Álvaro, Linhares, Formankiewicz & Waugh, 2022). The EnChroma lens cuts three such notches at wavelengths taken from a normal retina (Gómez-Robledo, Valero, Huertas, Martínez-Domingo & Hernández-Andrés, 2018). It makes colors look more different along the red–green axis, while discrimination thresholds change minimally (Somers et al., 2024). Two commercial lenses shifted discrimination thresholds in 51 individuals with red–green CVD. For both lenses, variation between observers exceeded the mean

effect (Patterson et al., 2022). Software filters start from the same population-average retina. They forward-project a stimulus onto an anomalous retina, either by the Brettel–Viénot–Mollon construction (Brettel, Viénot & Mollon, 1997) or by the parameterized Machado model (Machado, Oliveira & Fernandes, 2009). Daltonization inverts these models. It recolors an image, moving red–green differences onto the blue–yellow and lightness axes (Akalın, Aksoy, Top & Kara, 2025).

Individualized correction has been implemented at the retinal level. It tunes a cone-shift parameter to a user’s responses on plate-style or appearance-matching tests. That parameter can be estimated accurately, from opsin genotype or from cone spectral sensitivities measured in observers of known genotype (Deeb, 2005; Stockman & Sharpe, 2000). The retinal parameter predicts color discrimination imperfectly. Some anomalous trichromats with very small spectral separations discriminate better than their separation predicts (Neitz & Neitz, 2011). Across observers, the correlation between cone spectral separation and color discrimination is weaker than expected (Bosten, 2019). A retinal parameter fixes the input to the visual system. It does not describe the cortical representation that is built from that input and determines color perception. The remaining step is a filter derived from the user’s own cortical color representation.

Cortical activity patterns specify both which color was seen and where that color lies among the others. Cortical color processing is distributed and hierarchical (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals, and hue-selective responses to intermediate colors appear already in V1 (Parkes, Marsman, Oxley, Goulermas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). In hV4, a hue withheld from training can be reconstructed from the responses to its neighbors (Brouwer & Heeger, 2009). The relative positions of colors in these response patterns constitute a geometry. That geometry is shared across observers, and a color can be identified in one observer from the response patterns of others (Bannert & Bartels, 2025).

What has been measured of those response patterns in CVD is the strength of the

color signal. Anomalous trichromats perceive more red–green contrast than a cone-sensitivity account predicts (Boehm, MacLeod & Bosten, 2014). Deutan observers adapt to chromatic contrast more than the same account predicts (Basim, Goddard, Yang & Webster, 2025). Both results are consistent with a post-receptoral gain that magnifies the weakened L–M signal. Long-term adaptation to the altered input can produce such an adjustment (Webster, 2015). In V1, color responses in anomalous trichromats match the reduction their cone contrast predicts. In V2 and V3 they are indistinguishable from normal trichromats (Tregillus et al., 2021).

Color deficiency also changes where colors lie relative to one another. Anomalous trichromats still describe stimuli with all four hue primaries, but each primary lands on a different stimulus than it does for normal trichromats (Emery, Kuppuswamy Parthasarathy, Joyce & Webster, 2021). When dichromats judge how different pairs of colors are, red and green fall close together. The map recovered from those judgments has one axis where normal trichromats have two (Saysani, Corballis & Corballis, 2018). Protan and deutan observers keep both axes, but their map of the hue circle is dented at yellow and purple-blue. The size of that dent differs tenfold among observers who carry the same diagnosis (Ohkoba, Ishikawa, Hira, Ohtsuka & Ayama, 2021). All of these are judgments. A model of an anomalous trichromat’s color representation must be built from cortical measurements of the same geometry.

We take the individual’s own cortical color representation as the reference and invert it into a correction on the stimulus. Among the regions that carry this geometry, only hV4 predicts behavior on a trial-by-trial basis in a color judgment task (Bannert & Bartels, 2018). Our design adopts the premise that shifting an individual’s hV4 color geometry toward that of the healthy controls moves their color perception with it.

Three gaps remain. (*Gap 1*) The cortical color geometry of an individual with CVD remains to be described. Existing descriptions come from healthy observers and are reported at the group level (Brouwer & Heeger, 2009, 2013; Kuriki et al., 2015). (*Gap 2*) The distances between colors have not been measured in the cortical response in CVD. fMRI in CVD has reported response magnitude, including contrast-response

gain in anomalous trichromacy (Tregillus et al., 2021) and activation in dichromacy and achromatopsia (Rina, 2024). (*Gap 3*) A filter inverted from an individual’s own cortical representation remains underinvestigated. Existing designs invert retinal models.

We ask whether an individual’s own cortical color representation can be turned into a correction for that person. We characterize two adults with CVD, one deutan and one protan, against a reference group of healthy controls (see Methods).

A stimulus-space filter is warranted and tractable in one specific regime, defined by two conditions. The displayed colors must remain decodable from the cortical response. This preserved decodability gives the filter a signal to act on. The continuous arrangement of hues must be distorted, so that there is something to correct. Our questions follow this logic.

1. **Describe.** Does an individual’s cortical hue geometry in CVD carry structured distortion, with the individual hues still decodable and their continuous arrangement impaired?
2. **Summarize.** Can a few interpretable parameters capture that distortion, treated as a compact description rather than as physiological measurements?
3. **Correct.** Can a realizable, individualized filter be derived from those parameters?
4. **Validate.** Does the filter move the cortical representation closer to the healthy-control reference than a deployed accessibility filter does?

We expect the continuous arrangement of hues to be selectively impaired while the individual hues remain decodable. A few interpretable parameters should capture this distortion, and inverting them should yield a realizable, individualized correction.

Together, these steps yield a filter built for one individual. Its input is the cortical signal, its parameters are the fitted description, and its output is a stimulus shift. Each is defined at the level of the single observer, the level at which any correction must ultimately act. This study tests the feasibility of the inversion, namely whether an individual’s cortical color geometry can be turned into a realizable filter. It does not

test whether per-person correction outperforms a subtype average, which would require several individuals within one subtype.

2 Methods

The analysis proceeded in three stages, summarized in Figure 1C. First, we characterized each CVD participant’s cortical color representation with three neural analyses. Leave-one-run-out (LORO) decoding tested whether color classification remained intact (§2.8). Leave-one-color-out (LOCO) decoding tested whether continuous hue interpolation was impaired and yielded a per-hue vulnerability profile (§2.8). Procrustes disparity quantified how far each CVD hue geometry departed from the healthy-control reference (§2.9). Second, we fitted two classes of CVD distortion model by minimizing a combined neural and psychophysical loss. Parameters were selected by a pre-specified three-gate procedure (§2.10–§2.12). Finally, we inverted each fitted distortion numerically to obtain a per-subject stimulus-space correction filter (§2.14). Figure 3 summarizes the fitting and inversion procedure. Each filter was evaluated in a second session that combined fMRI with the two psychophysical tasks described in §2.3.

2.1 Participants

Thirteen volunteers were recruited and provided written informed consent. The study was approved by the Seoul National University Institutional Review Board (SNUIRB No. 2510/002-023) and conducted in accordance with the Declaration of Helsinki. All participants reported normal or corrected-to-normal visual acuity. Color vision was screened with the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants who identified 9 or more plates correctly were classified as healthy controls (HC, $n = 7$, 3 female, aged 20–27 years, mean 22.7 ± 2.5). Participants scoring below this threshold were classified as CVD. The classification plates of the same test distinguished the protan from the deutan subtype and indicated anomalous trichromacy rather than dichromacy in both. Two CVD participants completed both sessions, both male. The deutan participant was 24 years old and identified 5 of 14 plates correctly.

The protan participant was 25 years old and identified 7 of 14 plates correctly. Four further participants were excluded. Three did not complete the first scanning session. The fourth completed it but became unavailable before the filter-evaluation session. All CVD results are reported as single-case comparisons against the HC distribution with the Crawford and Howell (1998) modified t -test. No population-level inference is made.

2.2 fMRI stimuli and task

Eight chromatic stimuli were defined in CIE $L^*a^*b^*$ space at equal angular spacing in the a^*b^* plane, from 0° to 315° in 45° steps (Commission Internationale de L'Eclairage, 1986). All eight shared a lightness of $L^* = 75$ and a chroma of 40, and they spanned red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler at the same lightness with $a^* = b^* = 0$ served as a null condition. Isoluminance was not determined for individual participants. Stimuli were presented on a BOLDscreen 32 UHD monitor (Cambridge Research Systems) at 3840×2160 resolution, which combines a wide-gamut LED backlight with manufacturer-integrated calibration. The display was used at its factory calibration without further colorimetric verification in the scanner environment. Stimulus size is reported in display pixels throughout, since both presentation programs addressed their display in pixel units.

Stimuli were presented in a rapid serial visual presentation (RSVP) design to sustain attention and fixation. Each trial displayed one uniform chromatic disc, 500 pixels in diameter, for 1.5 s in a pseudo-random order optimized with Neurodesign (Durnez, Blair & Poldrack, 2018). A rapid serial letter stream ran at the center of the disc, and participants pressed a key on an MRI-compatible button box whenever the letter K appeared twice in succession. Inter-stimulus intervals were drawn from $\{3, 4.5, 6\}$ s. Each run comprised 72 events and lasted approximately 7 min, and each session comprised six runs. The experiment was implemented in PsychoPy 2022.2.5 (Peirce et al., 2019).

The schedule was optimized for detection power rather than for equal counts, so color counts varied within a run. Each chromatic stimulus occurred 5–9 times per run

and 38–47 times across the six runs, and the remaining 14–20 events per run presented the gray filler. Every participant received the same optimized schedule, so this allocation contributes a common component to all individual estimates. The general linear model estimated one amplitude per color and run regardless of how many trials contributed, so all downstream analyses operated on a balanced $6 \text{ (run)} \times 8 \text{ (color)}$ design.

2.3 Psychophysical tasks

All participants completed two psychophysical tasks on the day of the first scanning session. The first measured just-noticeable differences (JND) in hue discrimination, and the second was an eight-alternative forced-choice (8AFC) color-identification task. Both CVD participants repeated both tasks under each filter in the second session (§2.15). Both tasks ran outside the scanner under normal room lighting, on a built-in Liquid Retina XDR display set to the Apple XDR Display color preset, which was also the preset used to drive the in-scanner display.

Just-noticeable difference (JND). On each trial two chromatic discs appeared simultaneously to the left and right of fixation, and participants judged whether the two were the same or different. Each disc had a radius of 96 pixels in a 1024×768 window, their centers were separated by 260 pixels, and the side of the reference disc was randomized across trials. A fixation cross preceded each trial for 0.5s, and the discs remained on screen until the participant responded. The two hues were separated by a fraction t of the pair’s full separation, interpolated in CIE LCh , and t served as the staircase level. Two staircases per pair were interleaved from starting levels $t = 0.8$ and $t = 0.5$, both following a 1-up/1-down rule that converges on the 50% “different” point (Levitt, 1971). The step size was 0.15 for the first two reversals and 0.03 thereafter. A staircase terminated at eight reversals and at least 20 trials, extended by two further reversals whenever the last six reversal levels had an SD above 0.10. The threshold was the mean of the last six reversal levels obtained after the first two. The two discs always differed, so the procedure included no same-trials and the resulting threshold is not corrected for response bias.

Eight hue pairs were selected to span the S-cone axis and each subtype’s confusion axis. Yellow–purple lies along the S-cone axis, and blue–purple and red–orange flank it. Orange–yellow and yellow–green bracket the deutan confusion axis, and green–blue and cyan–magenta bracket the protan confusion axis. Red–cyan, separated by 180° , served as a control against ceiling effects. Pair selection was fixed on these grounds before any threshold was collected. For each pair we computed the JND ratio $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$, in which $\bar{\gamma}_{\text{HC}}$ denotes the HC mean threshold. Ratios above 1 indicate elevated hue-discrimination difficulty relative to HC. Because all participants were measured on the same display, the ratio is unaffected by any difference between that display and the one used in the scanner. These ratios entered the fitting loss described in §2.11.

8AFC color identification. Each trial began with a 1.0 s fixation, followed by a uniform chromatic disc of 500 pixels in diameter for 1.5 s and a 0.5 s blank interval. Participants then selected the presented hue from the eight scanner stimulus colors, displayed as a two-by-four grid of discs 100 pixels in radius at 240-pixel horizontal and vertical spacing, and navigated with the arrow keys. The positions of the eight options were reshuffled on every trial. A trial was recorded as a timeout if no response was given within 10 s, and no feedback followed. Each run comprised 64 trials, eight per color, in random order. Identification accuracy served as an independent psychophysical readout, held out from the fitting loss.

2.4 MRI acquisition and preprocessing

Imaging was performed on a Siemens 3T MAGNETOM Cima.X scanner with a 64-channel head coil. A whole-brain T1-weighted MPRAGE volume served as the anatomical reference. It was acquired sagittally at 1 mm isotropic resolution, with a repetition time of 2000 ms, an echo time of 2.36 ms, a field of view of 256 mm, a bandwidth of 220 Hz per pixel, and GRAPPA acceleration factor 2. Functional coverage was restricted to posterior occipital cortex with 24 interleaved oblique slices aligned perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009), spanning 48 mm. The functional sequence used a repetition time of 1.5 s, an echo time of 30 ms, a flip angle of

70°, and 2 mm isotropic voxels over a 192 mm field of view on a 96×80 matrix. Acceleration was GRAPPA factor 2 without multiband, the bandwidth was 1447 Hz per pixel, and phase encoding ran right to left. Each run yielded 288–292 volumes. A gradient-echo field map with echo times of 3.88 and 6.34 ms was acquired in each session and associated with the functional runs in the BIDS metadata.

Data from the first session were converted to BIDS format with the ezBIDS web service (Levitas et al., 2024), accessed in October 2025, which also defaced the anatomical image. The second session was converted with dcm2bids 3.2.0 (Boré, Guay, Bedetti, Meisler & GuenTher, 2024) in June 2026 and was not defaced. Both routes used `dcm2niix` for DICOM conversion. Preprocessing followed a custom pipeline rather than a standard container, and no confound regression was applied at any stage (Supplementary §S1). All steps used FSL 6.0.5.1 and FreeSurfer 7.2.0. The T1w image was skull-stripped with `bet2` (Smith, 2002), and functional images were registered to it by mutual-information coregistration initialized from the scanner header, using `mri_coreg` (Wells, Viola, Atsumi, Nakajima & Kikinis, 1996; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997). Normalization to the ICBM 152 Nonlinear Asymmetric 2009c template (MNI152NLin2009cAsym) used a twelve-parameter affine transform (FLIRT) followed by nonlinear warping (FNIRT) (Jenkinson, Bannister, Brady & Smith, 2002; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic BOLD data in MNI space. Each functional volume was resampled once by this composed transform, without slice-timing correction, motion realignment, susceptibility distortion correction, or spatial smoothing. Head motion was estimated separately by rigid-body registration of every volume to its run’s middle volume, using MCFLIRT (Jenkinson et al., 2002), and retained as a quality-control record. Mean framewise displacement was 0.32 ± 0.04 mm across the nine analyzed participants (HC 0.31 ± 0.04 , CVD 0.34 ± 0.05), with 16.2% of volumes exceeding 0.5 mm (Supplementary §S1). Because the primary analysis left head motion uncorrected, every neural endpoint was recomputed with the motion parameters and their temporal derivatives entered as nuisance regressors (Supplementary §S2).

2.5 ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the probabilistic atlas of visual topography of Wang, Mruczek, Arcaro and Kastner (2015), resampled to 2 mm and thresholded at a probability above 50%. The threshold was applied to each region independently, so a voxel assigned to more than one region was retained in each. Each region was then intersected with that participant’s BOLD brain mask. Mean voxel counts across the nine analyzed participants were 665 for V1, 458 for V2, 105 for V3, and 64 for hV4, with standard deviations of 209, 113, 19, and 18. No further functional voxel selection was applied.

Response amplitudes were estimated with a two-stage general linear model (Dale, 1999; Brouwer & Heeger, 2009, 2013). In the first stage a finite impulse response model with eight delays, spanning 0 to 10.5 s, was fitted to the concatenated runs without drift terms. Averaging the fitted responses over voxels and over the eight colors gave one hemodynamic response function per region and participant. In the second stage that function and its temporal derivative were convolved with per-color event sequences and fitted to each run separately by ordinary least squares, alongside a linear and a constant drift term for that run. The result was one amplitude for every voxel, color, and run, arranged as a $6 \times 8 \times V$ array per participant.

Amplitudes were aligned across runs by a within-subject Procrustes rotation (Gower, 1975). For each of runs 2 to 6, the 8×8 orthogonal matrix Q minimizing $\|X_1 - X_r Q\|_F$ was applied to that run’s color-by-voxel matrix X_r , where X_1 is run 1. Scaling was not permitted, so the rotation acts within the eight-dimensional space of color responses and leaves each voxel’s identity unchanged.

2.6 Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i^{\text{srn}} \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i^{\text{sr}}, S} \sum_{i=1}^N \|X_i - W_i^{\text{sr}} S\|_F^2 \quad \text{s.t.} \quad W_i^{\text{sr}\top} W_i^{\text{sr}} = I_k \quad (1)$$

The orthonormality constraint preserves distances and angles among the eight color patterns when they are mapped into the shared space. SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}}^{\text{sr}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. The reduced dimension was chosen by seven-fold leave-one-subject-out cross-validation, withholding one HC participant from SRM training on each fold. Candidate values from 2 to 6 were ranked on each fold by three quality metrics, namely RDM split-half reliability, cross-subject RDM correlation, and reconstruction error, and the value with the best mean rank was selected. This gave $k = 4$ for V1 and V2 and $k = 3$ for V3 and hV4.

2.7 Forward encoding model

Both classification (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a $F = 6$ -dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$, where $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k , with preferred hues spaced equally at 60° . Stacking the eight stimulus channel vectors row-wise gives the channel design matrix $C \in \mathbb{R}^{8 \times F}$. Decoding and encoding optimize different targets and therefore use different estimators of the channel-to-voxel weight matrix $W \in \mathbb{R}^{F \times V}$, which is distinct from the SRM mapping W^{sr} .

Decoding (stimulus hue from voxels). Decoding recovers the stimulus hue from voxel responses. We estimated W by ordinary least squares, taking the pseudoinverse of the channel design matrix C . For a held-out voxel pattern $\mathbf{x}$, channel activations were reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$. The decoded hue was the angle, among all 360 integer hues, whose basis vector correlated most strongly with $\hat{\mathbf{c}}$ (Brouwer & Heeger, 2009; Figure 2). For eight-way classification accuracy this angle was assigned to

its nearest stimulus hue. This correlation readout was carried over from the forward model and was subsequently compared against five alternative decoders, which is reported in Supplementary §S10.

Encoding (voxel responses from channels). Encoding predicts the voxel responses from the channels as $\hat{X} = CW$, where $\hat{X} \in \mathbb{R}^{8 \times V}$. We estimated W by ridge regression. A single penalty α was chosen for each fit by generalized cross-validation (Golub, Heath & Wahba, 1979), minimizing the GCV score averaged over the voxels of the region (Supplementary §S7). Regularization is warranted because the few stimuli and the correlated basis channels would otherwise overfit W and degrade prediction of held-out responses (Kay, Naselaris, Prenger & Gallant, 2008; Naselaris, Kay, Nishimoto & Gallant, 2011). This ridge prior is isotropic and imposes no spatial structure on the voxels.

2.8 Two decoding schemes: cross-run classification and cross-color interpolation

Two complementary cross-validation schemes applied the forward encoding model to characterize each participant’s hue processing. Leave-one-run-out (LORO) removed one run at a time. Every hue therefore remained in the training set, and the scheme tested the categorical representation of the eight hues, measured as color classification. Leave-one-color-out (LOCO) removed all six runs of one hue at a time. The held-out hue therefore never appeared in training, and the scheme tested the continuous representation, measured as hue interpolation.

Cross-run classification (LORO). Color classification was assessed by leave-one-run-out cross-validation in SRM-aligned space. For each of the six runs, W was estimated on the remaining five runs from mean-subtracted SRM projections, and the eight color patterns of the held-out run were decoded by the correlation readout of §2.7. Primary performance was the eight-way classification accuracy averaged across the six folds, against a chance level of 0.125.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject

generalization scores. The encoding model trained on each HC participant’s SRM-aligned data was applied to every other participant’s aligned patterns, yielding 21 unique HC-to-HC and 14 HC-to-CVD pairwise scores. Each participant contributes to several pairs, so these scores are not independent and the test is reported as a descriptive group-level check. Within-ROI single-case tests used the Crawford and Howell (1998) modified t -test, one-tailed and lower.

Cross-color interpolation (LOCO). Continuous hue interpolation was assessed by leave-one-color-out cross-validation. Each of the eight hues was held out in turn. W was estimated from the remaining seven hues across all six runs, pooled into a 42-sample design matrix, and the held-out hue was predicted by the same correlation readout.

The primary region for interpolation was hV4. Brouwer and Heeger (2009) reconstructed novel colors from V4 and VO1, the regions where their interpolation was strongest. LOCO was computed in all four regions. Crediting a region with interpolation required the healthy-control mean to exceed the color-label permutation null defined below. This criterion tests specificity to color identity rather than overall decodability. It also fixes where a CVD reduction is interpretable, since the comparison rests on controls who interpolate above the null. Single-case CVD comparisons are reported for regions meeting the criterion. Values for all four regions are given in the Results.

Adjacent accuracy. The primary cross-color decoding metric was adjacent accuracy, the proportion of predictions falling within 45° of the target hue. This tolerance corresponds to one hue step. Values were averaged across the six runs of each test color and range from 0 to 1. The readout selects among all 360 integer hues, so a prediction drawn uniformly from that output space falls within tolerance on 91 of 360 draws, giving a chance level of 0.25. Exact accuracy, which requires the prediction to fall in the correct one of eight 45° bins, has a chance level of 0.125. Healthy-control performance was tested against an empirical null of 1,000 random color-label permutations rather than against the analytic chance level. Single-case HC–CVD

comparisons at hV4 used the Crawford and Howell (1998) modified t -test, one-tailed and lower.

Vulnerability profile. The eight-dimensional vector of per-hue adjacent-accuracy values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. For group-level model fit, we report Spearman ρ between the observed $\mathbf{v}$ and the $\mathbf{v}$ predicted by the CVD distortion model (see the Candidate CVD distortion models section). Per-hue group comparisons (Figure 4C) used the Crawford and Howell (1998) modified t -test, one-tailed and uncorrected. Effect sizes are reported as the case-control index $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}}$, which equals $t\sqrt{(n+1)/n}$ (Supplementary §S19).

2.9 Representational geometry

Procrustes disparity measures how far a participant’s whole representational geometry lies from the healthy-control reference in SRM-aligned space. Disparity was the Procrustes distance between a participant’s SRM-aligned $8 \times k$ pattern and that reference. Both patterns were centered on their column means and scaled to unit Frobenius norm. Disparity was then the residual Frobenius norm after the optimal orthogonal rotation.

The HC disparity distribution was estimated by a leave-one-HC-out procedure. Each HC participant was compared to the mean of the remaining six, which keeps that participant out of its own reference. Individual CVD disparity was compared to this distribution by the Crawford and Howell (1998) modified t -test, one-tailed upper at $\alpha = 0.05$. The direction follows the prediction that CVD geometry lies farther from the HC reference than HC geometry does.

Two estimates of the distribution were computed (Supplementary §S7). In the first, the shared SRM space is trained on all seven HC, so each HC reference is in-sample while the projected CVD participant is out-of-sample. In the second, the space is retrained on six HC at each fold, and the held-out HC is projected by SVD on the same terms as each CVD participant. Each CVD participant receives the mean of its seven fold disparities, whereas each HC participant contributes the single value from

the fold that held it out. The control distribution therefore retains fold-to-fold variance that the CVD statistic averages out, which makes the comparison conservative.

The two estimates serve different purposes. Every other stage of the pipeline operates in the all-HC space, so that space defines the representation the filter is fitted in. Testing whether a participant departs from the healthy-control distribution requires the held-out participant to be projected on the same terms as the CVD participants, which the symmetric estimate provides. We report both and treat the symmetric estimate as the inferential test (table S4).

2.10 Candidate CVD distortion models

Two candidate model classes were evaluated. Each parameterizes the angular distortion $\delta\theta(\theta)$ that maps the displayed hue angle θ to the angle perceived by a CVD observer.

Candidate Model 1: retinal-plus-gain (R+C).. The prevailing account attributes CVD color distortion to a cone-spectral shift at the retinal level (Machado et al., 2009), optionally augmented by a cortical compensation gain g (Boehm et al., 2014; Tregillus et al., 2021):

$$\delta\theta_{\text{RC}}(\theta; g) = (2 - g) \delta\theta_{\text{Mach}}(\theta) \quad (2)$$

where $\delta\theta_{\text{Mach}}$ is the per-hue angular shift predicted by Machado simulation applied to Stockman & Sharpe LMS cone fundamentals (Stockman & Sharpe, 2000). The gain g parameterizes cortical compensation: at $g = 1$ the cortical stage leaves the retinal distortion unchanged; at $g = 2$ it fully cancels the retinal shift; at $g > 2$ cortical processing reverses the direction of the retinal shift, moving the perceived hue past the undistorted color. The confusion axis is the color-space direction along which CVD observers confound hues with one another. Since $\delta\theta_{\text{Mach}}$ lies entirely along that axis, R+C carries a single free parameter and displaces colors along one fixed direction.

Candidate Model 2: two-component cortical (2-comp). The second model parameterizes the distortion as a sum of two cosine components, one on the

S-cone axis and one on each subtype’s confusion axis:

$$\delta\theta(\theta; \beta_s, \beta_c) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}}) \quad (3)$$

where θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan), derived from Stockman & Sharpe cone fundamentals (Stockman & Sharpe, 2000). β_s parameterizes distortion along the S-cone axis ($\theta = 90^\circ$) and β_c along the individual confusion axis. Each component has a fixed direction and a free amplitude, so the model spans distortions in the plane defined by those two axes. Both signs of β_c are unconstrained. We constrain $\beta_s \geq 0$ as a weak directional prior that fixes only the sign, not the magnitude, of the S-cone-axis term, consistent with reports that CVD perceptual errors concentrate near the S-cone confusion loci Emery et al., 2021. Model adequacy was assessed by the procedure in §2.12; results are reported in §3.5.

2.11 Inverse fitting

Per-subject distortion parameters (β_s, β_c) were estimated by finding the distortion that, when applied to HC-derived neural representations, best reproduces the observed CVD responses. Three loss families served as the fitting objective. A specific instance of a family — for example, the aggregate JND ratio or the ΔRDM loss at a single ROI — is a *loss atom*; candidate objectives are combinations of atoms, selected as described in §2.12.

Psychophysical JND ratio (L_γ). A distortion changes the perceived angular separation of a hue pair, and a pair that is compressed becomes harder to discriminate. We modeled the predicted threshold for a pair as $\hat{\gamma} = \bar{\gamma}_{\text{HC}} (d_{\text{phys}}/d_{\text{perc}})$, where d_{phys} is the separation of the two hues on the display, d_{perc} is their separation after the candidate distortion of Eq. 3, and $\bar{\gamma}_{\text{HC}}$ is the HC mean threshold for that pair. The loss is the squared difference between the predicted and the observed CVD threshold, standardized by the HC standard deviation s_{HC} for that pair and summed over the pairs entering the atom:

$$L_\gamma = \sum_p \left(\frac{\hat{\gamma}_p - \gamma_{\text{CVD},p}}{s_{\text{HC},p}} \right)^2 \quad (4)$$

The atoms available to the fitting were single-pair terms and one aggregate term. For the deutan participant the single-pair candidates were orange–yellow and yellow–green, which bracket that subtype’s confusion axis, and yellow–purple, which lies on the S-cone axis. For the protan participant the single-pair candidate was green–blue, which brackets the protan confusion axis. The aggregate term γ_{all} sums over all eight tested pairs. Which atom entered each participant’s objective was decided by the procedure in §2.12. The reported ratio $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$ (§2.3) places the same thresholds on a common scale and serves reporting rather than fitting.

Per-pair RDM difference (ΔRDM).. Procrustes disparity summarizes the distance between two geometries as a single value after optimal rotation. It reports how far a participant’s geometry lies from the HC reference, and the optimal rotation removes the information about which colors are displaced. Inverting the distortion requires that information. We therefore expressed the same comparison as a difference of representational dissimilarity matrices, which retains the displacement of every color pair. For each participant and ROI the RDM was the 8×8 matrix of pairwise correlation distances ($1 - \text{Pearson } r$) among the reduced color patterns of that region, and the per-pair difference was $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$, in which $\overline{\text{RDM}}_{\text{HC}}$ is the mean RDM over the seven HC participants.

Representational ΔRDM cosine (L_{RDM}).. For a candidate distortion, a simulated difference $\Delta\text{RDM}_{\text{sim}}$ is obtained by applying $\delta\theta$ to the HC mean RDM and comparing it to the observed ΔRDM . Each distorted hue is assigned to the nearest of the eight 45° bins, so $\Delta\text{RDM}_{\text{sim}}$ is built by reindexing the entries of the HC mean RDM. A candidate distortion therefore acts as a permutation of the eight hues, and distortions that map to the same permutation are indistinguishable to this term. The loss is the cosine dissimilarity between the observed and simulated 28-element upper-triangle vectors:

$$L_{\text{RDM}} = 1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}}) \quad (5)$$

Because the loss is a cosine, the magnitude of ΔRDM does not enter.

The loss computes ΔRDM in a PCA-reduced space ($K = 6$). Across the 28

pairwise entries this estimate agreed with the SRM-aligned estimate used for disparity at V1, V2 and hV4 ($r = 0.77\text{--}0.89$), and less closely at V3 ($r = 0.39\text{--}0.58$), so the reduction does not determine the pattern the loss fits.

hV4 LOCO voxel-prediction (L_{LOCO}). This atom scores how well a candidate distortion accounts for the CVD participant’s own held-out responses at hV4. Weights were fitted within that participant. For each held-out color c , ridge weights W_c were estimated from the remaining seven colors across all six runs, with the penalty chosen by generalized cross-validation as in §2.7. The candidate distortion relabels each stimulus by its perceived hue, so the held-out response is predicted as $\hat{Y}_c = \mathbf{c}(\theta_c + \delta\theta_c)^\top W_c$. The fit for that color is the Pearson correlation ρ_c between $\hat{Y}_c$ and the participant’s run-averaged pattern at color c . The loss averages the complement over the eight colors:

$$L_{\text{LOCO}} = \frac{1}{8} \sum_{c=1}^8 (1 - \rho_c) \quad (6)$$

Fitting the weights within participant keeps the prediction in that participant’s voxel space, since the number of voxels differs across participants and HC-trained weights do not transfer. This atom was available to every candidate objective, and the selection procedure of §2.12 decided whether it entered.

Composite loss. The atoms differ in scale, since L_γ is built from threshold ratios while L_{RDM} and L_{LOCO} are bounded dissimilarities. Each atom was therefore evaluated over the full parameter grid and standardized to zero mean and unit standard deviation across that grid. The composite loss is the sum of the standardized atoms divided by $\sqrt{n_a}$, where n_a is the number of atoms. Standardization sets the relative weight of the atoms and makes any constant rescaling of an individual atom immaterial.

Grid search. Parameters were searched by exhaustive grid at 2° resolution, with β_s over $[0^\circ, 50^\circ]$ (26 values) and β_c over $[-50^\circ, 50^\circ]$ (51 values), giving 1,326 cells. For R+C (Eq. 2), g was searched over $[0, 3]$ in steps of 0.05 (61 values). Per-subject loss combinations and parameter selection are described in §2.12.

2.12 Parameter selection

Per-subject loss combinations and model class were selected by a three-gate procedure. Two resampling schemes over the seven HC references supplied the inputs, a 5-train/2-test resample repeated $N = 300$ times and a strict 7-fold leave-one-HC-out. Both used the same loss atoms.

Gate 1: Separation precondition. Admission required the CVD loss value to differ from the HC leave-one-out distribution by a Cohen’s d of magnitude 0.5 or greater, in either direction. Atoms below that threshold were excluded from combination.

Gate 2: Boundary saturation. Among admitted combinations, those in which at least half of the resample solutions saturated the grid boundary were rejected, since boundary saturation indicates model misspecification. Combinations were also rejected when the recovered parameters collapsed across resamples, defined as an interquartile range above 50° or a sign reversal between the training and test argmin exceeding 5° .

Gate 3: Held-out test-loss. Surviving combinations were ranked by the median composite test-loss $\bar{L}_{\text{test}}$ across the $N = 300$ resamples. For each resample the composite loss was evaluated at the training argmin on the two held-out HC and z-normalized relative to the training grid, so a lower $\bar{L}_{\text{test}}$ indicates generalization to unseen HC references. Ties were broken by the test-loss interquartile range, which measures the stability of that generalization estimate. These two quantities were the only ranking criteria.

Further quantities were recorded for each candidate outside the ranking. The held-out neural loss $\bar{L}_{\text{RDM}}^{\text{LOO}}$ from the strict 7-fold leave-one-out evaluates a candidate’s neural atoms with the psychophysical term excluded. It therefore isolates what the neural terms alone contribute to the fit. Parameter stability was summarized by the interquartile range of the recovered $(\hat{\beta}_s, \hat{\beta}_c)$, by the fraction of resamples recovering the modal 45° bin, and by the strict 7-fold parameter range. The HC false-positive rate is likewise reported descriptively in §3.7.

2.13 Identifiability and recovery

Four pre-specified checks were run on each production candidate before any parameter was reported.

Parameter recovery. This test establishes whether a known distortion can be recovered at the present sample size and noise level. Synthetic CVD responses were generated at a known ground truth by passing that distortion through each control participant’s encoder. Noise matched that participant’s residual structure, using the top 20 principal components of its spatial covariance and an AR(1) correlation of 0.3 across runs. The grid fit was re-run on each synthetic dataset, giving $n = 140$ refits per candidate from seven control encoders and twenty noise realizations. Recovery was summarized by f_{10° , the fraction of refits within 10° of ground truth on both axes, against a pass criterion of $f_{10^\circ} \geq 0.5$ with $|\text{bias}| < 10^\circ$. The refit repeats the grid search rather than the full selection procedure, so it measures recoverability at a fixed loss combination.

Origin-recovery null. This test records the spread the procedure returns when no distortion is present, which sets its per-axis uncertainty floor. The same synthesis was run with ground truth at $(0^\circ, 0^\circ)$ and each control participant’s real JND. It returns no p -value and was excluded from the test pool.

HC pseudo-CVD specificity. This test asks whether a control participant placed in the CVD position yields a comparable estimate. Each HC participant’s real amplitudes were substituted into the CVD slot in turn, giving a seven-member null and a one-sided rank-distance percentile.

Color-label permutation. This test asks whether the estimate depends on which color evoked which response. The eight color labels of the real CVD amplitudes were permuted $N = 1000$ times, with the same permutation applied across ROIs and the HC pool left unchanged, giving a one-sided p_{perm} .

The three test-bearing checks across the two production candidates gave six tests, corrected together under the Benjamini–Hochberg false-discovery rate ($\alpha = 0.05$). Outcomes are reported in §3.7.

2.14 Stimulus-space filter

The correction filter for each participant is the pre-image of the fitted transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the eight displayed hues θ_k we solved $T(\tilde{\theta}_k) = \theta_k$ for the input angle $\tilde{\theta}_k$, which gives a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. Under the model, a CVD observer viewing $\tilde{\theta}_k$ then obtains the cortical hue representation that a healthy control observer obtains when viewing θ_k .

Each pre-image was located by Brent root-finding on the residual $T(\theta) - \theta_k$, searched within $\pm 60^\circ$ of the target and refined to a bracket tolerance of 10^{-9} degrees. The transform is not guaranteed to be invertible for arbitrary parameters, so every solution was verified by forward-substitution. All eight hues resolved to within 10^{-3} degrees of their target for both deployed parameter sets, and the procedure rejects a parameter set that leaves any hue unresolved.

2.15 Filter evaluation

Both CVD participants completed a second scanning session that compared the individualized filter and a deployed macOS accessibility filter against the unfiltered Session-1 baseline. The individualized filter was the per-subject 2-component stimulus-space pre-image, with parameters frozen from the main analysis and therefore out-of-sample. Acquisition, comparator, run-count, and single-case-inference details are given in Supplementary §6.

On the second-session data we recomputed, within each condition, the primary Session-1 interpolation metric (LOCO adjacent accuracy, §2.8), LORO classification, SRM Procrustes disparity into the HC shared space, and the representational dissimilarity matrices. As a secondary encoding index we also computed the forward-tuning correlation, the Pearson r between the predicted and the observed voxel pattern at each held-out hue, averaged over the eight hues. It indexes the encoding fit rather than the decoding accuracy that carries the comparison, and is reported as corroboration only.

Each condition’s neural index was reported as a standardized single-case effect

size (Cohen’s d against the HC distribution) rather than an inferential p , because the grand-mean permutation null is biased for this design. In both CVD participants, the JND and 8AFC tasks (§2.3) were repeated under each filter, each compared to the unfiltered Session-1 baseline and to the HC distribution. This interleaved, cross-session psychophysical comparison is descriptive.

2.16 Reproducibility

All analyses were conducted in Python 3.10 with numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0 (Pedregosa et al., 2011), and BrainIAK 0.11. Random seeds were fixed for every permutation test and bootstrap. Most analyses used seed 42, while the parameter-recovery synthesis used 31337 and the color-label permutation used 27182, and each script records its own value. Analysis code and preprocessing scripts are available at https://github.com/Transconnectome/colorBlind_analysis. Anonymized preprocessed data are available on reasonable request to the corresponding author.

3 Results

3.1 All eight colors remain decodable across ROIs in both CVD cases

A corrective stimulus-space filter is valid only if the target hues remain decodable from the visual cortex of a CVD individual. We verified this precondition using leave-one-run-out (LORO) cross-validation of the forward encoding model in SRM-aligned space.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 4A). Their accuracies also sat inside the healthy-control distribution at every ROI, with no single-case comparison reaching significance (Crawford–Howell, eight tests, all $p \geq 0.095$; at hV4 both participants gave $d_{cc} = -0.95$, $p = 0.407$; Supplementary §S19). A forward encoding model trained on the healthy controls and applied to a CVD participant decoded that participant’s held-out runs at 0.432 on average, far above the 0.125 chance level ($t(7) = 6.51$, $p < 0.001$), so the control-derived

mapping transfers to the CVD data. The 8 scanner stimuli therefore remain individually decodable at the cortical level in both CVD participants, in line with prior work on CVD above-threshold identification (Boehm et al., 2014; Bosten, 2019).

3.2 Hue interpolation is reduced at hV4 in both CVD cases

We next tested whether continuous hue interpolation fails in CVD, using leave-one-color-out (LOCO) decoding of the forward encoding model. Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy 0.47 ± 0.05 , mean $\pm$ SEM over $n = 6$ HC, sub-07 excluded at hV4 for insufficient voxels; $p = 0.008$ under 1,000 per-subject label permutations). V1–V3 did not significantly exceed chance even though the colors remained decodable across runs (V1 $p = 0.164$; V2 and V3 below chance). This is consistent with hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Adjacent accuracy at hV4 reached 0.13 in the protan participant and 0.25 in the deutan participant, against a healthy-control mean of 0.47 and a chance level of 0.25 (Figure 4B). The protan value fell below the HC distribution (Crawford & Howell $t = -2.91$, $p = 0.017$, $d_{cc} = -3.14$). The deutan value carried a comparably large effect at $p = 0.063$ ($d_{cc} = -1.99$, $t = -1.84$), in line with the power of a single-case test against $n = 6$ controls.

Descriptively, the reduction was not uniform across hues. Both participants showed zero adjacent accuracy at hV4 for the S-cone intermediate colors blue, purple, and magenta (Figure 4C). Per-hue single-case tests (Crawford & Howell, one-tailed, uncorrected and exploratory) reached significance at no individual hue against the $n = 6$ HC distribution. These per-hue accuracies form each participant’s hue vulnerability profile $\mathbf{v} \in [0, 1]^8$.

3.3 Geometric deviation localizes to a distinct ROI in each CVD case

To characterize the geometric basis of the LOCO impairment, we compared the representational geometry of CVD and HC participants in a shared SRM-aligned space.

Procrustes disparity indexes how far a participant’s whole geometry lies from the HC reference.

The ROI of elevated disparity differs between the two CVD participants, V2 in the deutan participant and V1 in the protan participant (Figure 5B). The protan participant showed elevated disparity specifically at V1, under both the common HC space ($p = 0.007$) and the symmetric LOSO reference ($p = 0.045$), with no elevation at V2, V3, or hV4. The deutan participant showed elevated disparity at V2 in the common HC space ($p = 0.040$), which did not survive the symmetric LOSO reference ($p = 0.116$, $d_{cc} = 1.42$), with no elevation at V1, V3, or hV4. The direction and magnitude of the deutan effect were preserved under the LOSO reference while the confidence interval widened, so the two estimators differ in precision rather than in sign. Distinguishing an attenuated effect from an absent one at this ROI requires a larger cohort. Full test statistics and effect sizes (t , d_{cc}) for both estimators are reported in table S4.

The two cases differ in where the deviation lies and in how strongly it is supported. Because the deviation is not shared, a family-level correction cannot serve both. In each participant the continuous arrangement of hues is displaced in a direction and magnitude specific to that individual, which is what a personalized correction is required to capture. We next fit this distortion with a low-dimensional model — using each participant’s neural representational geometry and psychophysical hue-discrimination thresholds (§3.5) — and invert the fitted model into a per-subject stimulus-space correction (§3.8).

3.4 The retinal-plus-gain model is structurally insufficient

The R+C model is structurally insufficient for both CVD participants, though the supporting evidence differs by case. For the deutan participant, 100% of resample solutions saturated the grid boundary ($g = 3.0$), exceeding the 50% boundary-saturation gate (Section 2.12) and rejecting the model as misspecified (Wilson & Collins, 2019). For the protan participant, saturation was lower (41%, $g = 2.95$) and did not by itself trip that gate; here the model is excluded instead by its fitted overcompensation ($g > 2$,

below) and its degree-of-freedom deficit (below), and it is in any case dominated on held-out loss by the 2-component fit ($\bar{L}_{\text{test}} = -0.86$ vs -1.54 ; Section 3.5).

A gain $g > 2$ implies that cortical processing reverses the direction of the retinal shift past the undistorted hue (Section 2.10). Both participants scored below the Ishihara criterion (the deutan participant: 5/14 plates correct; the protan participant: 7/14), confirming residual CVD under standard viewing conditions. A fitted $g > 2$ is therefore internally inconsistent with confirmed CVD status: the model claims overcompensation where the participants demonstrably show impairment. Therefore, the saturated boundary is a model failure, not a valid parameter estimate.

The R+C family also carries a degree-of-freedom restriction. The function $\delta\theta = (2 - g) \cdot \delta\theta_{\text{Mach}}$ displaces hues along the fixed confusion axis alone, for every value of g . It therefore omits displacement away from that axis, including along the S-cone axis of the 2-component model. R+C is excluded as a filter candidate.

3.5 A common cortical model fits both CVD cases

The 2-component cortical model fit both participants using the same functional form. The deutan and protan cases differ in $(\hat{\beta}_s, \hat{\beta}_c)$, not in model structure. We fit the two-component parameters (β_s, β_c) for each participant and inverted the fitted transform to a stimulus-space filter.

For the deutan participant ($\theta_{\text{conf}} = 150^\circ$), the selected loss combination $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$. The held-out test-loss was $\bar{L}_{\text{test}} = -2.36$ (IQR = 2.15), lower than the competing β_s -dominant candidate $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -10^\circ)$, which reached only $\bar{L}_{\text{test}} = -1.14$ and was therefore not adopted. The HC-resample parameter IQR was $(8^\circ, 2^\circ)$. The strict 7-fold LOO $\hat{\beta}_c$ range was $[-46^\circ, -38^\circ]$; all values are negative, confirming a stable sign across held-out reference sets.

For the protan participant ($\theta_{\text{conf}} = 16^\circ$), the selected loss combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$. The held-out test-loss was $\bar{L}_{\text{test}} = -1.54$ (IQR = 1.42), nearly tied with its closest grid competitor ($\bar{L}_{\text{test}} = -1.52$); the choice among neighboring 2-component cells was therefore weakly determined, although both

improved on the alternative retinal-plus-cortical model class scored on the same composite ($\bar{L}_{\text{test}} = -0.86$), locating the advantage in the model class rather than the precise grid cell. The HC-resample parameter IQR was $(0^\circ, 0^\circ)$, and 87.7% of resamples recovered the same 45° bin. The strict 7-fold LOO IQR was $(0^\circ, 0^\circ)$. However, this stability is specific to the PCA-basis loss: the argmin shifts under the alternative SRM-basis reduction, so the recovered position is metric-dependent.

For neither participant did the LOCO loss family enter the selected combination; both winning combinations were JND-plus- Δ RDM.

The ROI supplying the RDM atom was fixed differently in the two participants, and we report that asymmetry explicitly. For the deutan participant the ROI was a free term in the search: all four retinotopic ROIs entered the combination grid, and $\gamma_{\text{OY}} + L_{\text{RDM}}^{(\text{V2})}$ ranked first among the 25 combinations passing the boundary-saturation gate (Section 2.12), at $\bar{L}_{\text{test}} = -2.36$ against -2.20 for the next combination and -1.72 for the best V1 combination. For the protan participant the ROI was pre-specified as V1 and was not searched. Completing the ROI axis post hoc under the same selection rule selects V2 instead ($\bar{L}_{\text{test}} = -2.20$ vs. -1.54 for V1), but every ROI that passes the gate — V1, V2, and V1+V4 — returns the identical estimate $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$, so the filter carried into the second session is unaffected. The same invariance holds for the deutan participant within its selected psychophysical atom, where V2 and V3 both return $(6^\circ, -42^\circ)$ and dropping the RDM atom entirely returns $(16^\circ, -44^\circ)$. The ROI choice therefore locates the neural loss term but does not determine the fitted distortion.

On held-out folds, both fitted distortions reproduced the observed CVD geometry, beating the no-correction baseline on all 7 folds and ranking in the top 5–8% of grid combinations. The HC noise ceiling — the best fit achievable given HC data noise — was estimated from split-half HC amplitude reliability (Lage-Castellanos, Valente, Formisano & De Martino, 2019) and is conservative because CVD amplitude noise is excluded. Relative to this ceiling, the fits reached 52% (the deutan participant) and 67% (the protan participant) of the way from baseline to ceiling (held-out losses in table S9).

3.6 Neural data identifies what psychophysics alone cannot

Fitting psychophysical (γ) and neural (RDM) loss atoms independently reveals three benefits of using the neural component.

First, the neural term captures a protan cortical direction that the psychophysical loss cannot resolve. The protan participant’s psychophysical-only fit yielded $\hat{\beta}_c \approx +4^\circ$ and did not beat the no-correction baseline ($\Delta L = +0.01$, 3/7 folds). The RDM term captures a signal that the psychophysical loss cannot detect for this participant.

Second, the neural term stabilizes the argmin for the deutan participant. The psychophysical-only and combined fits share the same argmin ($6^\circ, -42^\circ$), confirming that the psychophysical atom independently locates the deutan direction; the neural-only fit independently corroborates this direction with an estimate of $\hat{\beta}_c = -26^\circ$. Adding the RDM atom reduced the boundary saturation rate from 23% to 9.3%, sharpening the argmin without shifting it.

Third, the neural term enhances parameter stability by reducing its IQR in both participants. For the deutan participant, adding the neural term reduced the HC-resample IQR from $(18^\circ, 6^\circ)$ to $(8^\circ, 2^\circ)$ (PCA) and $(10^\circ, 4^\circ)$ (SRM). For the protan participant, the IQR narrowed from $(6^\circ, 4^\circ)$ to $(0^\circ, 0^\circ)$ (PCA, deterministic) and $(0^\circ, 2^\circ)$ (SRM).

3.7 Identifiability: mechanism class recoverable, per-axis magnitude not

These fits recover the distortion’s mechanism class and the direction of its dominant axis, but not its precise per-axis magnitudes. Parameter recovery fails across all pre-specified identifiability checks (0/6 significant after FDR correction, BH $\alpha = 0.05$; Supplementary §S16). The voxel-level parameter-recovery test yielded $f_{10^\circ} < 0.30$ for both production candidates. The non-dominant axes ($|\hat{\beta}_s| \leq 6^\circ$ for both participants) fall below the effective parameter uncertainty of $\sim 20^\circ$ – 25° and are not recoverable (Supplementary §S16). The dominant axis ($|\hat{\beta}_c| = 42^\circ$ for the deutan participant) exceeds this uncertainty and is partially recoverable: when CVD responses are simulated from a known distortion and re-fitted, the recovered $\hat{\beta}_c$ deviates from the

true value by 4.7° on average (Supplementary §S16).

Consistent with the partial recoverability of the dominant axis, the sign of $\hat{\beta}_c$ — the distortion direction — is stable across held-out reference sets (Section 3.5). The deutan participant consistently returns $\hat{\beta}_c < 0$ across all LOO folds and resamples. The protan participant consistently returns $\hat{\beta}_c > 0$ under the PCA-basis loss; sign stability under the SRM-basis loss was not verified (Supplementary §S16). The distortion direction is recoverable even when its magnitude is not.

The loss landscape shows where this magnitude indeterminacy lies. Although each participant’s selected fit is the global minimum on the full data (Figure 6), the per-resample argmins do not all concentrate there. For the protan participant, 87.7% recur at the selected fit, but a secondary basin near $(32^\circ, 0^\circ)$ marks a metric-dependent alternative. For the deutan participant, they disperse along the β_c axis, leaving its magnitude loosely constrained.

3.8 Per-subject stimulus-space filter

The per-subject filter $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ is the pre-image of the fitted 2-component transform (Section 2.14).

For the deutan participant ($\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -42^\circ$, $\theta_{\text{conf}} = 150^\circ$), the dominant $\hat{\beta}_c$ term produces a large-amplitude correction that varies asymmetrically across hues (mean $|\delta\theta| = 26.3^\circ$; Figure 7).

For the protan participant ($\hat{\beta}_s = 2^\circ$, $\hat{\beta}_c = +24^\circ$, $\theta_{\text{conf}} = 16^\circ$), the smaller $\hat{\beta}_c$ produces a more targeted correction concentrated near the protan confusion axis (mean $|\delta\theta| = 16.2^\circ$; Figure 7).

The two fits differ in the magnitude and direction of the dominant confusion-axis term ($\hat{\beta}_c = -42^\circ$ for the deutan participant vs. $+24^\circ$ for the protan participant).

3.9 Psychophysical and neural filter evaluation

Filter evaluation. We evaluated the two filters in a second session in both CVD participants, using the same two neural readouts as the first session: hue interpolation (hV4 LOCO accuracy) and representational geometry (SRM-aligned

disparity and RDM). We report, in turn, the psychophysical outcome, the color-classification precondition, and the two neural readouts.

Psychophysics. Both filters improved hue-discrimination thresholds and color identification in the deutan participant. The baseline deficit (JND mean $|z| = 2.24$; the orange–yellow, yellow–green, and yellow–purple pairs each exceeded the HC maximum) was removed by both the deployed ($|z| = 0.85$) and the individualized ($|z| = 0.78$) filter, neither leaving a significant deviant pair. 8AFC color identification rose from 0.81 (95% CI [0.70, 0.89]) to 0.97 ([0.89, 0.99]) under both (Wilson score interval, $n = 64$). The two filters could not be ranked (Wilcoxon $p = 0.84$). The protan participant had no significant baseline deviant pair, though one (green–blue, a protan confusion-axis pair) was trending ($p = 0.070$). Under the deployed filter green–blue became significantly deviant ($p = 0.003$) and a second pair appeared (cyan–magenta $p = 0.001$), whereas the individualized filter normalized green–blue ($p = 0.44$) and produced no deviant pair; mean $|JND-HC|$ (in HC SD units) was lower under the individualized filter (no-filter $0.90 \approx$ individualized $0.93 \ll$ deployed 1.78). 8AFC color identification, already at ceiling without a filter (1.00, 95% CI [0.94, 1.00]; Wilson score interval, $n = 64$), was preserved by the individualized filter (0.98, [0.92, 1.00]) but reduced by the deployed filter (0.86, [0.75, 0.92]). Of the two filters, only the individualized filter introduced no new significant deviation in either participant. Whether it is superior to the deployed filter remains to be established with more participants.

Colors remained decodable. All eight hues remained decodable under both filters in both participants. LORO eight-way classification accuracy stayed well above the 0.125 chance level at every ROI and in every condition (all cells ≥ 0.50 ; HC 0.71–0.77; Supplementary Table S14). The color signal was intact in both.

Interpolation. For color interpolation (LOCO decoding), the two participants gave opposite results. In the deutan participant, the individualized filter partially improved adjacent accuracy (no-filter $0.23 \rightarrow$ individualized 0.31 ; HC 0.46 ± 0.11), whereas the deployed filter did not (deployed 0.25). Both conditions stayed below the HC level ($d_{cc} = -1.35$ individualized, -1.94 deployed), and the individualized value

alone exceeded the 0.25 chance level. In the protan participant the trend reversed. Interpolation accuracy under the individualized filter was below the unfiltered baseline (no-filter 0.14, deployed 0.19, individualized 0.06; individualized $d_{cc} = -3.70$), the lowest of the three conditions, and all three fell below the 0.25 chance level. The individualized filter thus improved interpolation in the deutan participant only, and the effect did not replicate in the protan participant. A secondary forward-tuning encoding index corroborated the deutan improvement (hV4 $\rho = +0.18$ vs. HC $+0.21$; deployed $\rho = -0.39$) but did not separate the conditions in the protan participant (Fig. S2).

Geometry. Neither filter fully restored the representational geometry to the HC level, and the direction of the filter effect reversed between the two participants (Fig. 8B,C,E,F). In the deutan participant the unfiltered baseline was the closest to HC of the three conditions, while both filters increased the difference from HC (V2 SRM disparity no-filter 0.72 $\rightarrow$ deployed 0.84 / individualized 0.77, HC 0.49; V2 RDM similarity to HC no-filter 0.57 $\rightarrow$ deployed 0.15 / individualized -0.13 , HC-self 0.50). In the protan participant the unfiltered baseline was the most distorted of the three, while both filters reduced the difference from HC (V1 SRM disparity no-filter 0.76 $\rightarrow$ deployed 0.63 / individualized 0.62, HC 0.45; V1 RDM similarity no-filter 0.25 $\rightarrow$ deployed 0.49 / individualized 0.30, HC-self 0.66). This protan recovery was equivalent between the two filters on SRM disparity and larger under the deployed filter on RDM similarity. It was therefore not specific to the individualized filter.

4 Discussion

We derived a per-person color-correction filter for each of two CVD participants. Each filter inverts a model of one participant’s cortical color representation, fitted to that participant’s own neural and psychophysical measurements. That representation carried a structured geometric distortion rather than a uniform attenuation of the chromatic signal. Continuous-hue interpolation fell to chance in both participants, and representational disparity from the healthy-control (HC) geometry exceeded the disparity measured among controls themselves, while all eight hues remained decodable

at every early-visual ROI. We modeled the distortion as a hue rotation about two axes of color space, the subtype confusion axis and the S-cone axis. Both axes were fixed before fitting (Methods §2.11). The confusion axis is each subtype’s classical confusion-line direction, taken from cone fundamentals. The S-cone axis is the direction in which S-cone excitation varies, and the interpolation deficit concentrated along it. Inverting the fitted model gives a replacement color for any target hue. We applied it at the eight stimulus hues. In a two-person evaluation, the individualized filter removed the deutan participant’s baseline threshold deficit and normalized a trending confusion-axis pair in the protan participant, while every other pair stayed within the HC range. The deployed accessibility filter instead rendered two pairs deviant in the protan participant. On cortical hue interpolation the filter’s effect varied by participant, with adjacent-hue accuracy increasing in the deutan participant and decreasing in the protan participant. Establishing an advantage over the deployed filter in a larger sample is the immediate next test.

4.1 A geometric distortion of cortical color representation

In both CVD participants the deficit took the form of a structured distortion of cortical color geometry, localized to a different area in each. Two complementary measurements resolve it. The representational dissimilarity matrix (RDM) encodes the pairwise distance structure among stimuli within a cortical region (Kriegeskorte, Mur & Bandettini, 2008), and it identifies which of those distances depart from the HC geometry. Disparity from HC was elevated at V2 in the deutan participant and at V1 in the protan participant (Section 3.3). Leave-one-color-out decoding identifies where interpolation across the continuous hue circle falls to chance. In the HC cohort, hV4 alone supported above-chance interpolation, replicating Brouwer and Heeger (2009). Both CVD participants fell at or below chance at hV4, most severely at the S-cone intermediate hues blue, purple, and magenta (Section 3.2). This pattern aligns with prior characterizations of CVD as a multidimensional deformation of color space rather than a uniform attenuation (Boehm et al., 2014; Ohkoba et al., 2021; Emery et al.,

2021). That the two participants’ distortions localized to different areas is consistent with the perceptual case. Among anomalous trichromats, between-observer variability in hue scaling is 3.4 times the within-observer variability (Emery et al., 2021).

4.2 A neurally grounded, individualizable correction filter

Inverting each participant’s fitted two-component model assigns every hue a single replacement color (Methods §2.14). That replacement is the correction filter, the displayed hue that the participant’s own cortical transformation maps onto the intended color. Across the eight hues the filter applied a mean rotation of $|\delta\theta| = 26.3^\circ$ for the deutan participant and 16.2° for the protan participant.

To our knowledge, this is the first color-correction filter derived from an individual’s cortical color representation rather than from a retinal model. Its parameters come entirely from that individual’s measured cortical color geometry, whereas population-average corrections apply a fixed subtype-average spectral shift. Prior inversions of cortical encoding models set the magnitude of response in a chosen unit or region rather than the arrangement of colors within it (Bashivan, Kar & DiCarlo, 2019; Shinkle & Lescroart, 2025).

The neural term made three distinct contributions to the fit. First, it resolved a confusion-axis rotation that the psychophysical loss alone left near zero. For the protan participant, the psychophysical-only fit selected $\hat{\beta}_c \approx +4^\circ$, and its held-out loss equalled that of applying no filter ($\Delta L = +0.01$, improving on it in 3 of 7 folds). The combined fit recovered $+24^\circ$. Second, it reinforced a solution the psychophysical loss had already located. For the deutan participant the psychophysical-only and combined fits shared the same argmin ($6^\circ, -42^\circ$). The neural term reduced the boundary-saturation rate from 23% to 9.3%. The neural-only fit returned the same sign of $\hat{\beta}_c$ (-26°), corroborating the psychophysical estimate. Third, it narrowed the HC-resample interquartile range of the argmin in both participants, from $(18^\circ, 6^\circ)$ to $(8^\circ, 2^\circ)$ in the deutan participant and from $(6^\circ, 4^\circ)$ to $(0^\circ, 0^\circ)$ in the protan participant.

The two fitted distortions diverge, with $\hat{\beta}_c = -42^\circ$ in the deutan participant

against $+24^\circ$ in the protan participant. The sign of $\hat{\beta}_c$ remained stable when the HC reference set was resampled. Leaving out each of the seven HC reference participants in turn confined $\hat{\beta}_c$ to $[-46^\circ, -38^\circ]$ in the deutan participant and returned an identical argmin on every fold in the protan participant. The fitted signs record displacement at two different anchor directions in a shared hue frame, and the two confusion axes lie 134° apart ($\theta_{\text{conf}} = 150^\circ$ and 16°).

The per-axis magnitudes are not separately identifiable, with 0 of 6 recovery checks surviving FDR correction. What the fits recover is the sign of the dominant confusion-axis term. In the protan participant that sign is basis-dependent. The fit used an RDM loss computed in a PCA-reduced space, and computing the same loss in the SRM-aligned space moves the argmin (Supplementary §6). With one participant per subtype, individual and subtype differences remain confounded.

The retinal-plus-gain alternative carries a single degree of freedom, and its cortical gain rescales displacement along the confusion axis alone (Appendix A). It therefore omits displacement away from that axis, including the S-cone intermediate hues where the interpolation deficit concentrated (Section 3.2). Fitting the class registered this misspecification. In both participants the gain settled above 2, which implies cortical overshoot past the undistorted hue and is inconsistent with their confirmed residual deficit (Section 3.4). The single-axis restriction precludes an exact per-person correction, and the correction must instead be derived from the cortical measurement.

4.3 Filter evaluation

The individualized filter left every hue-discrimination pair within the HC range in both participants, whereas the deployed filter rendered two pairs deviant in the protan participant. In the deutan participant both filters removed the baseline threshold deficit. In the protan participant the individualized filter normalized the trending confusion-axis pair green–blue, one of the two pairs the deployed filter left deviant. Color identification accuracy followed the same ordering. Both filters raised it in the deutan participant, whereas in the protan participant it was preserved under the

individualized filter and reduced under the deployed filter. This psychophysical advantage appeared in the protan participant only.

The filter’s effect on the neural readouts varied by participant and by measure. Adjacent accuracy on hue interpolation separated the two participants, rising in the deutan participant ($0.23 \rightarrow 0.31$) and falling in the protan participant ($0.14 \rightarrow 0.06$). In each participant the early-visual geometry shifted in the direction opposite to interpolation. In the deutan participant both filters moved the geometry away from the HC reference. In the protan participant both filters moved the geometry toward the HC reference. So, that recovery was not specific to the individualized filter. SRM disparity was lower under the individualized filter in the deutan participant and nearly equal between filters in the protan participant. RDM similarity to HC was higher under the deployed filter in both participants. The geometry of both participants remained displaced from the HC reference.

Early-visual geometry and hV4 interpolation lie at different levels of the visual hierarchy and can vary independently. These neural readouts remain inconclusive in a two-case sample. Replication in more individuals within each subtype, on a refined neural endpoint, is required to establish whether either pattern generalizes.

4.4 Limitations

The CVD sample is $N = 2$, one participant per subtype. Subtype was assigned from Ishihara classification plates rather than anomaloscopy, so the confusion-axis angle assumed for each participant carries the uncertainty of that assignment. Single-case statistics (Crawford & Howell, 1998) permit individual-level inference, but population-level claims about deutan or protan subtypes require replication with more participants. Separating a per-person correction from a subtype-average one requires testing several individuals *within* a single subtype. It also requires grading severity on a continuous scale. The plates classify subtype but do not quantify severity. An anomaloscope, which measures the red–green mixture an observer accepts as a match to a reference yellow, would supply both that grading and an independent check on the

subtype assignment. The HC reference cohort is likewise small at seven controls. The HC $\|\hat{\beta}\|$ leave-one-out distribution is therefore reported descriptively, as a reference for the magnitude of the CVD estimates rather than as a significance test.

Two of the reported estimates depend on analysis choices. The deutan V2 disparity elevation is significant in the common HC space and falls to a non-significant trend under the symmetric leave-one-subject-out control, whereas the protan V1 elevation is significant under both (Section 3.3). We also report $\hat{\beta}_s$ and $\hat{\beta}_c$ as single values without confidence intervals, because the six runs per participant are too few for stable bootstrap resampling.

The stimulus set was confined to a single locus of constant lightness and chroma ($L^* = 75$, chroma = 40). The correction is therefore untested for stimuli that vary in lightness or chroma, including natural scenes. Extending it to those stimuli requires refitting the model at additional levels of both dimensions. Matching lightness in CIE $L^*a^*b^*$ also equates the stimuli for the standard observer rather than for an individual observer. We measured neither the luminance of the displayed stimuli nor each participant’s luminous efficiency. Including both as measured covariates would separate chromatic from luminance contributions to the cortical geometry.

The two neural terms in the fitting objective rest on different quantities. L_{LOCO} scores voxel-pattern prediction at hV4 using weights fitted within the participant. L_{RDM} scores the direction of a 28-element vector of pairwise distances at V1 or V2, referenced to the control mean. The interpolation term separated both CVD participants from the control distribution by a large margin. However, it did not enter either selected combination under the three-gate procedure. The same disconnect appears at evaluation, where adjacent accuracy, representational disparity, and RDM similarity to controls moved in different directions under the same filter (Section 3.9). A comprehensive neural index would encompass both interpolation and representational geometry. Sharing that index between the fit and the evaluation is a prerequisite for the joint neural endpoint that a larger study would need.

4.5 Conclusion

We derived a stimulus-space color correction for each of two CVD participants by inverting a model fitted to that participant’s own cortical color representation. All displayed hues remained decodable in both participants, whereas the continuous hue geometry departed from the HC reference, at a different cortical area in each. In these two cases the deficit took the form of a hue rotation about two axes of color space rather than a uniform attenuation of the chromatic signal. The fitted parameters differed between the two participants, and inverting each model assigned every displayed hue a single replacement color. Each correction is therefore a per-person transform, not a population-average retinal one.

Larger and systematic studies can now quantify the cortical color distortions of CVD and yield a class of individualized, cortically grounded filters. In the present two-person evaluation the correction’s effect varied by participant and by measure. Whether it provides a generalizable improvement in color perception is what those studies would establish.

CRedit authorship contribution statement

[FILL AT SUBMISSION] Assign the 14 CRedit roles to each author, e.g. Jinil Kim: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing — original draft, Writing — review & editing. Confirm the final author list first.

Declaration of competing interests

[FILL AT SUBMISSION] State whether any author holds a competing financial or personal interest. Use the Elsevier template sentence when none applies.

Acknowledgements

[FILL AT SUBMISSION] Funding sources with grant numbers, scanner facility, and non-author contributions.

Data availability

[FILL AT SUBMISSION] Name the repository holding the preprocessed amplitudes and the analysis code, with the access conditions and a DOI or URL. Reconcile with the corresponding sentence in Methods.

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Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section 3.5). Here we compare it against the retinal-family alternative on the same held-out composite criterion (§2.12), and give the structural reasons the retinal-family class is inadequate for the stimulus-space filter task.

A.1 Retinal-plus-cortical (R+C) fit on the composite criterion

The retinal-plus-cortical model applies the subtype Machado cone-shift $\delta\theta_{\text{Mach}}$ Machado et al., 2009 scaled by a single cortical-gain parameter g ($\delta\theta_{\text{RC}} = (2 - g) \delta\theta_{\text{Mach}}$; $g = 1$ retinal-only, $g = 2$ exact cortical cancellation; Tregillus et al., 2021). Scored on the same held-out composite as the 2-component model, R+C failed for both participants: for the deutan participant every resample saturated the gain boundary ($g \rightarrow 3.0$, 100%), so the fit was rejected as misspecified; for the protan participant the best fit ($g = 2.95$) generalized worse than the adopted 2-component model ($\bar{L}_{\text{test}} = -0.86$ vs. -1.54). Full fits are in Supplementary Table S9 and Section 3.4. A gain $g > 2$ implies cortical overshoot past the undistorted hue, inconsistent with the participants’ confirmed residual CVD, so the saturated estimate is a model failure rather than a parameter value.

A.2 Inverse-mapping (pre-image) behavior

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The filter reported in Section 3.8 is that pre-image, evaluated at the eight stimulus hues.

A.3 Why the cortical gain cannot recover the distortion

The cortical gain g rescales the retinal shift uniformly, so for every value of g the R+C family displaces hues along the fixed confusion axis alone. It cannot generate the 2-component model’s S-cone-axis term $\beta_s \cos(\theta - 90^\circ)$, a direction that lies 60° from the deutan confusion axis and 74° from the protan one. The observed interpolation deficit

concentrates at the S-cone intermediate hues (Section 3.2), a displacement pattern this family cannot represent. The restriction is a property of the model class and holds across fitting criteria.

6 Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated by rigid-body registration of every volume to its run’s middle volume (FSL MCFLIRT) and summarized as framewise displacement (Power, Barnes, Snyder, Schlaggar & Petersen, 2012), with rotations converted to displacement on a 50 mm sphere. Across the nine analyzed participants mean framewise displacement was 0.32 ± 0.04 mm (HC 0.31 ± 0.04 , CVD 0.34 ± 0.05), and 16.2% of volumes exceeded 0.5 mm. These parameters served as a quality-control record only, and the analyzed images were resampled once from native BOLD space into MNI space. A gradient-echo field map was acquired in each session and linked to the functional runs through the BIDS `IntendedFor` field. The analyzed pipeline did not consume it, so the functional data retain susceptibility-induced distortion along the right–left phase-encoding direction. Slow drift was modelled with per-run linear regressors in the general linear model, which constituted the entire nuisance model. Temporal filtering was not applied.

S2. Uncorrected acquisition artifacts

The functional series were acquired with a limited field of view covering occipital cortex. Registration therefore used header-initialized mutual information, which tolerates limited coverage and requires no white-matter boundary. Slice timing, rigid-body motion, and susceptibility distortion were all left uncorrected, and each volume was resampled once by the composed native-to-MNI transform.

Motion sensitivity analysis. Every neural endpoint was recomputed with the six motion parameters and their temporal derivatives added to the second-level design matrix. A control analysis added the same twelve regressors after circularly shifting each within run. The shift preserves the autocorrelation and spectrum of the regressors while destroying their temporal alignment with the data, and therefore separates the variance inflation caused by adding twelve regressors from the removal of motion-aligned variance.

The protan V1 elevation survives motion regression and strengthens under it. The deutan V2 value, already non-significant under the leave-one-subject-out reference, falls further. The shifted control retains the deutan elevation, so the attenuation under motion regression does not follow from the loss of degrees of freedom or from collinearity among the added regressors.

Split-half RDM reliability, averaged over the four regions, gives the same ordering. Taking the difference between the motion-regression arm and the shifted control as the net effect of removing motion-aligned variance, the deutan participant loses 0.338, which is 2.5 times the next largest value in the cohort (0.134), whereas the protan participant is unaffected (+0.002).

Head motion was not locked to the stimulus sequence in any participant. Regressing the motion parameters on the eight-color design matrix and comparing the resulting R^2 to a null built from circularly shifted onsets gave z between -0.07 and $+0.08$ and p between 0.34 and 0.63 across the ten participants.

Susceptibility distortion. Distortion displaces signal along the phase-encoding direction, which ran right to left. Within a participant it is constant across conditions, so it cannot generate condition-specific differences among the eight colors. Its cost is spatial, and it reduces the precision with which atlas-defined regions are assigned to functional voxels.

Coregistration stability. Registration of the occipital BOLD volume to the whole-head anatomical image was probed in two ways. Across the runs of a single session, where the fitted transform should be identical, the mean pairwise displacement of the solution ranged from 0.9 to 4.2 mm depending on participant and session. Removing facial voxels from the anatomical image, which leaves the brain unchanged, moved the solution by 1.9 mm in the deutan participant and 9.4 mm in the protan participant. The mutual-information optimum is therefore shallow when the field of view is restricted to occipital cortex. The transform is fixed within a run, so it cannot produce differences among the eight colors that a run contributes. Registration variability instead enters as run-to-run noise, which averaging over the six runs

attenuates.

Limits. Motion regression removes variance temporally aligned with the estimated parameters. It does not reconstruct the data that volume realignment would have produced, so this analysis bounds the contribution of motion-aligned variance rather than establishing what a fully corrected pipeline would yield. Realignment and distortion correction of limited-FOV series are both nontrivial, since rigid-body estimation is less constrained when little of the head is imaged. Acquisitions of this type would benefit from correction validated for partial coverage.

S3. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S4. Quality Control

ROI coverage averaged 84.3% (SD 21.7%) across the ten scanned participants, and 99.6% of ROI voxels carried reliable stimulus-evoked responses. Coverage was computed as the intersection of the Wang atlas ROIs (V1, V2, V3, hV4) with each participant’s BOLD brain mask in MNI space. Coverage reached 30.8% in sub-07, the lowest value in the sample, and all participants entered the downstream analyses.

S5. Mean Activation Analysis

Across 20 comparisons of activation-level metrics between HC and CVD, effect sizes spanned $|d| = 0.04$ to 0.93 with all $p \geq 0.094$. The metrics were mean activation magnitude (mean $|\beta|$), signal-to-noise ratio, color modulation depth, run-to-run reliability, and spatial variance, each evaluated at V1, V2, V3 and hV4 by Welch’s t -test on the amplitudes before SRM alignment. Both CVD cases fell inside the HC range on every metric (Crawford–Howell, all $p > 0.09$). Voxel-level color selectivity was substantial in both groups (one-way ANOVA across voxels, $F > 4$, $p < 0.001$ in V1–V3).

Activation metrics and SRM disparity varied independently across participants, with Pearson r from -0.29 to $+0.51$ over 20 tests and all $p \geq 0.130$. The HC–CVD difference therefore resides in the representational pattern rather than in response amplitude. The cortical color response retains its strength in CVD while its geometry departs from the control reference.

S6. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined by leave-one-subject-out cross-validation over the candidate values 2 to 6 (Cunningham & Yu, 2014). On each of the seven folds, one HC participant was withheld from SRM training and three quality metrics were computed for every candidate. RDM split-half reliability was the Spearman correlation between representational dissimilarity matrices built from the odd runs 1, 3, and 5 and the even runs 2, 4, and 6 of the withheld participant in SRM space. The remaining two metrics were the cross-subject RDM correlation and the reconstruction error. Candidates were ranked within each fold and metric, and the candidate with the best mean rank was selected. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S7. Generalized Cross-Validation for Ridge Regression

This penalty applies to the encoding direction alone. Hue decoding uses the unregularized pseudoinverse, because a ridge penalty on the channel-to-voxel weights degrades the decoding readout (§S8). The regularization parameter α for ridge regression was selected by generalized cross-validation (Golub et al., 1979), which approximates leave-one-out cross-validation at a fraction of its cost. We implemented it

directly from the singular value decomposition of the channel design matrix, searching the grid $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$. A single α was selected per fit by minimizing the GCV score:

$$\text{GCV}(\alpha) = \frac{\overline{\|Y - \hat{Y}\|_2^2}}{(1 - \text{tr}(H)/n)^2} \quad (7)$$

where C denotes the channel design matrix, Y the observed responses of all voxels in the region, $\hat{Y}$ the ridge prediction, $H = C(C^T C + \alpha I)^{-1} C^T$ the hat matrix, and n the number of training samples. The squared residuals are averaged over both samples and voxels, so the selected penalty is shared across the voxels of a region.

S8. Cross-Validation Procedures

Leave-one-run-out (LORO).. LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B , C) were held out as the test set, W was trained on the remaining five runs. The held-out run’s color patterns were then decoded by nearest-neighbor matching in channel-output space using the OLS pseudoinverse weights (Methods §2.7).

Leave-one-color-out (LOCO).. LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors. The held-out color was then decoded from voxel responses using the OLS pseudoinverse weights, yielding the adjacent-accuracy vulnerability profile. The complementary voxel-response prediction (ridge-GCV encoding) enters only the fitting loss L_{LOCO} , not the decoding metric (Methods §2.7).

Cross-run alignment and leakage control. The run-level Procrustes alignment (Methods, ROI definition and response estimation) maps runs 2–6 onto a fixed run-1 reference by an orthogonal rotation that uses no stimulus labels. Because the alignment target is a single fixed run rather than a jointly fitted shared space, the

held-out run or color does not define the alignment frame. Re-estimating the alignment inside each fold raised forward-encoding LORO accuracy from 0.545 to 0.578. Leakage in the fixed-reference pipeline would predict the opposite direction, so the reported estimate is not inflated by the alignment step. That variant also replaces the run-1 reference with the training-run mean, so it bounds the leakage question rather than isolating the nesting itself.

Leave-one-subject-out (LOSO)... LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants’ shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S9. Evaluation Metrics

Color decoding metrics. Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90° , expected from a uniform random predictor on a circle), and (2) eight-way classification accuracy, obtained by assigning the decoded hue to the nearest stimulus hue, equivalently a $\pm 22.5^\circ$ bin around each of the eight colors (chance = $12.5\% = 1/8$).

Voxel prediction metrics. Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical significance. Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S10. Comparison with alternative decoders

The forward encoding model was adopted for its interpolation behaviour rather than for its classification accuracy. We compared six decoders on the same SRM-aligned amplitudes under both cross-validation schemes. Classification separates the decoders

in one direction and interpolation separates them in the other, and the interpolation result is the one the present analyses depend on.

Linear discriminant analysis classified held-out runs most accurately at every ROI (table S2). Its healthy-control mean reached 0.878 at V1 against a chance level of 0.125, with the support vector machine next at 0.777 and the forward encoding model at 0.542. The ordering held at V2, V3 and hV4. Four of the six decoders exceeded chance at every ROI, so eight-way classification is recoverable by several readouts and does not by itself favour any one of them.

Interpolation to a held-out color reverses the ordering at hV4, the ROI on which the main analyses rest. The forward encoding model reached an adjacent accuracy of 0.470 there, against a chance level of 0.25 for its continuous-hue output (table S3). It is the only decoder whose healthy-control mean exceeds its own chance level, and it does so at V1 (0.360), V2 (0.283) and hV4, with the widest margin at hV4. The classifiers stay below their 0.375 chance level at every ROI, and ridge and kernel ridge stay below 0.25 at every ROI. The analytic chance level compares a decoder against a uniform prediction. The main-text criterion is stricter, since the color-label permutation null preserves the covariance among voxels, and hV4 is the region that meets it (§3.2). This pattern follows from the model form: the forward encoding model represents a hue as activation over six tuned channels and can therefore evaluate a hue it never saw during training, whereas a classifier trained on seven discrete labels has no output unit for the eighth.

The correlation readout entered this comparison as the incumbent rather than as its outcome. It was implemented for the Phase-1 forward model in January 2026 and specified in the analysis plan in February 2026, both before the runs reported here. The comparison therefore verifies that the readout was not the limiting factor, and the wording in the main text reflects that scope.

Two rows require care. Kernel ridge regression returned an adjacent accuracy of exactly 0.000 in every participant, ROI and alignment, and the multilayer perceptron returned exactly the chance value with zero variance across participants, which is the

signature of a constant prediction. Neither is treated as a competitive score. Three further variants are outside this table: the ensemble decoders were retired from the pipeline, the two hybrid decoders were run for interpolation only, and four alternative readouts of the forward encoding model vary the readout rather than the decoder.

S11. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-HC-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i} , yielding disparity scores $d(HC_i, R_{-i})$ and $d(CVD_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

Common-space vs. symmetric-LOSO estimates. The geometric-disparity tests (§3.3) are reported primarily in the common HC-trained SRM space, in which the seven HC train the shared space and each subject of interest is projected in; because the held-out HC reference is in-sample whereas the projected CVD participant is out-of-sample, we additionally computed a symmetric leave-one-subject-out (LOSO) estimate in which every HC and CVD subject alike is projected by SVD into a space trained on the remaining six HC (Methods, §2.9). Both use the Crawford & Howell, 1998 one-tailed single-case t ($df = 6$). Table S4 reports both estimates for all ROIs. The protan participant's V1 elevation is significant under both estimators; the deutan participant's V2 elevation is significant in the common space and attenuates to a non-significant trend under LOSO.

S12. Validity of the geometric comparison

Holding the shared projection fixed is required for the color-label permutation to carry power. Re-estimating the projection at each permutation lets the re-fit absorb the

shuffle. We tested this on the healthy controls, in whom color structure is established. Under re-estimation the procedure detected 0 of 7 controls at V1, V2 and V3, and 0 of 6 at hV4, with mean permutation z near zero in every ROI. Freezing the projection recovered detection, reaching 5 of 7 at V3 and 2 to 3 of 7 elsewhere (table S5). Every color-specificity result below therefore uses the frozen projection.

Both CVD participants carried color-specific geometry, at different ROIs. The deutan participant reached $p = .002$ at V2, the lowest V2 value among the nine participants, and $p = .024$ at V3. The protan participant reached $p = .001$ at V3 and $p = .013$ at V2, while V1 returned $p = .758$. Across the 35 participant-by-ROI cells, 16 fell below .05 against a chance expectation of 1.8, and 7 survived Benjamini–Hochberg correction over all 35 cells (table S6). Two of the surviving cells belong to the CVD participants, the deutan V2 cell ($q = .018$) and the protan V3 cell ($q = .012$). V3 carried color specificity most broadly, with 5 of its 9 cells surviving correction.

Motion regression broadened the pattern rather than removing it. The count of cells surviving correction rose from 7 to 15, and the deutan V2 cell held at $q = .025$ while the protan V3 cell held at $q = .012$. Two readings of this increase remain open. Motion may contribute label-independent variance that masks color structure, or the twelve added regressors may reshape the residual variance in a way that favours the observed correspondence. The circular-shift control that separates these accounts was applied to the disparity endpoint (S2) and remains to be extended to the permutation reported here.

Disparity and color specificity address different questions. Disparity measures the distance between two geometries after optimal rotation and stays invariant to how the colors map onto one another. The permutation asks whether the identity mapping between a participant’s colors and the reference colors outperforms a shuffled mapping. A participant can therefore sit far from the reference in disparity while the identity mapping remains the best correspondence available, or sit close to it while a shifted correspondence fits better.

The protan participant’s V1 correspondence is displaced by one hue step.

Rotating the color labels by 45° lowered Procrustes disparity from 1.037 to 0.788, a gain of 24.0%. We referred that gain to the distribution of shift gains across the seven healthy controls ($3.5 \pm 5.9\%$), which are computed under the same minimum-over-eight rule and therefore carry the same selection bias. The protan gain exceeded that distribution (Crawford–Howell $t = 3.22$, $p = .009$, $d_{cc} = 3.44$), and the rotated geometry sat below the control mean disparity of 0.839 ± 0.087 . The same rotation raised the second-order RSA similarity from $\rho \approx 0.00$ to $\rho = +0.52$, against a control mean of 0.45, exceeding its own selection-bias null ($z = +5.02$, $p = .002$, $p_{adj} = .032$). Motion regression preserved both effects (disparity gain 19.2%, $t = 2.43$, $p = .025$). The deutan participant showed the identity mapping as optimal at V1, V2 and V3.

The evidence supports the existence of a color rearrangement that restores control-level similarity in the protan participant at V1. Which rearrangement it is remains open. Under the SRM basis a 225° shift ($+0.50$) sits close to the 315° optimum ($+0.52$), and under the PCA basis the optimum falls at 135° ($p = .048$, which does not survive correction).

S13. Alignment-independent checks on the disparity measure

Disparity tracks the same subject-level quantity when the alignment method changes. We recomputed each participant’s distance from the healthy-control mean with two procedures that use no shared response model, then correlated the result with the SRM disparity used in the main analyses. Crossnobis distance in native voxel space agreed at $r = 0.632$ pooled across ROIs ($p < .001$), and PCA distance agreed at $r = 0.780$ ($p < .001$). Agreement was strongest at V1 and V2, the two ROIs carrying the single-case elevations (table S7).

Cross-validated Mahalanobis distance reproduces the ordering without any alignment step. Following Walther et al. (2016), we computed an 8×8 crossnobis distance matrix per participant in native voxel space, cross-validated over the 15 run pairs, with Ledoit–Wolf shrinkage of the noise covariance. Control-to-control RDM similarity exceeded control-to-CVD similarity at V1 by 0.104 (permutation $p = .120$),

and the three remaining ROIs gave differences below 0.05. The convergent correlation with SRM disparity reached $r = 0.833$ at V1 ($p = .005$) and $r = 0.733$ at V2 ($p = .025$).

A second alignment algorithm gives the same subject ordering. We replaced SRM with PCA at matched dimensionality, with and without a subsequent canonical-correlation alignment, and computed Procrustes disparity over all participant pairs. Group-level effects were small under both variants, with Hedges' g from -0.13 to $+0.40$ under PCA and from -0.13 to $+0.16$ under PCA-CCA. Pairwise alignment fits each pair independently and therefore carries more noise than a jointly fitted shared space, which accounts for the size of these effects. Subject-level agreement with SRM disparity was substantial under both (PCA pooled $r = 0.780$, $p < .001$; PCA-CCA pooled $r = 0.544$, $p < .001$).

The shared space reconstructs the CVD data at least as well as the control data, which places reconstruction quality outside the set of explanations for the elevated disparity. Under the symmetric leave-one-subject-out framework, variance explained was higher in CVD than in controls at all four ROIs, and the V2 difference was the largest (controls 0.331, CVD 0.416, $g = -1.26$). Variance explained and disparity varied independently across participants (pooled $r = -0.214$, $p = .211$), so the two indices track separate properties. High reconstruction quality together with high disparity describes a strong signal held in a different arrangement.

These checks establish that disparity is a property of the data rather than of the alignment procedure. They concern the measurement, and whether the deviation is specific to color is treated in S12. Group-level contrasts at this sample size carry wide intervals, so the argument rests on the subject-level agreement between three distances computed under three different alignment procedures.

S14. Comparison with Retinal-Family Distortion Models

The retinal-family (R+C) model was evaluated alongside the 2-component model under the same grid-search and held-out composite criterion (§2.12).

Model. The retinal-plus-cortical (R+C) model is defined in §2.10: the subtype

Machado cone-shift $\delta\theta_{\text{Mach}}$ Machado et al., 2009, fixed by the confusion axis, scaled by a single cortical-gain parameter g ($\delta\theta_{\text{RC}} = (2 - g) \delta\theta_{\text{Mach}}$; $g = 1$ is retinal-only, $g = 2$ exact cortical cancellation; Tregillus et al., 2021). It was fit by the same grid-search and held-out composite criterion as the 2-component model (§2.12), so the two model classes are directly comparable on $\bar{L}_{\text{test}}$.

Fits (Table S9). The retinal-plus-cortical model reached the gain boundary in both participants. For the deutan participant, 100% of resample solutions saturated at $g \rightarrow 3.0$, and the fit was rejected at the boundary-saturation gate as model misspecification (§3.4). For the protan participant, 41% of solutions saturated at $g = 2.95$, which passed that gate, and the held-out composite favoured the 2-component fit ($\bar{L}_{\text{test}} = -1.54$ against -0.86). A gain above 2 implies cortical overshoot past the undistorted hue, which the participants’ confirmed residual CVD excludes. The single-axis degree-of-freedom deficit accounts for this behaviour (§3.4).

Relation to ΔRDM and LOCO. Fitting a retinal-family model directly against ΔRDM reproduced the same boundary behaviour in both participants. For the adopted 2-component model, the ΔRDM cosine L_{RDM} enters the selected composite loss (§2.12). The forward-encoding LOCO family remains reserved for diagnostic decoding and entered neither adopted combination.

S15. HC leave-one-out magnitude-anchor distribution

Both CVD estimates fall inside the corresponding HC range, which places this section’s role as a descriptive magnitude anchor rather than a test of CVD-versus-HC specificity. The deutan estimate is $\|\hat{\beta}\| = 42.4^\circ$ against an HC range of 30.5° to 58.1° (mean 49.1°). The protan estimate is 24.1° against 23.4° to 55.5° (mean 35.7°), close to the lower bound of that range. Parameter magnitude alone therefore places both CVD cases within the HC distribution.

The production loss differs between participants, the deutan combination being $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ and the protan combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$, so each anchor was computed under that participant’s own loss. Each of the $n = 7$ HC participants was refit as a

pseudo-CVD case with the remaining HC pool as reference.

S16. Identifiability checks

Effective parameter uncertainty is approximately 20° on β_s and 25° on β_c under the PCA-basis loss, read from the median and interquartile range of the origin-recovery null (Test 2a). The production argmin therefore functions as a descriptive embedding of the CVD-specific distortion direction rather than as a point estimate with physiological magnitude. Four pre-specified checks establish this bound and delimit the claim. Test 1 assesses voxel-level parameter recovery, Test 2a measures the algorithm’s noise floor, and Tests 2b and 2c assess specificity to CVD data. Each test was run on the two production candidates (S08-robust, S09-primary), using PCA-basis voxel synthesis (spatial PCA(20) on SRM-projected amplitudes) as the feature space.

Benjamini–Hochberg correction across the six test-bearing checks (2 candidates $\times$ 3 tests, $\alpha = 0.05$) returned no result below threshold.

Test 1 — Voxel-level parameter recovery. Synthetic fMRI data were generated at the production argmin (PCA-basis; 7 HC donors $\times$ 20 noise draws = 140 samples per candidate) with GT-consistent JND: psychophysical thresholds were scaled to match the distortion magnitude at the ground-truth (β_s, β_c) . The same fitting pipeline was re-run on each synthetic sample. Pass criterion: fraction within 10° on both axes ($f_{10^\circ} \geq 0.5$ and $|\text{bias}| < 10^\circ$ on both axes).

Test 2a — Origin-recovery null. Synthetic fMRI data were generated at $\text{GT} = (0, 0)$ using real HC JND (no synthetic psychophysical signal; PCA-basis). At the null, real HC JND carries no CVD-consistent signal, making this a noise floor measurement free of synthesis design contamination. Pass criterion: $|\text{bias}| < 5^\circ$ on both axes.

Test 2b — HC pseudo-CVD specificity. Each HC participant’s real amplitude data were substituted as a fake CVD participant (7 carriers per candidate, deterministic). The production CVD fit distance from the origin was ranked among the 7 HC pseudo-CVD distances ($\text{rank}_{\text{dist}}$, one-sided high). Pass criterion: rank = 1.0 (CVD

fit exceeds all HC distances). This test is descriptive only and was not used as a selection criterion.

Test 2c — Color-label permutation. CVD trial labels were randomly shuffled ($N = 1000$ permutations); HC data were held fixed. Pass criterion: $p_{\text{perm}} < 0.05$ (one-sided, lower = better).

FDR summary and basis caveat. After FDR correction across 6 tests (2 candidates $\times$ 3 main tests; BH $\alpha = 0.05$), no test reaches significance. The production argmin is treated as a descriptive embedding of the CVD-specific distortion direction throughout.

All results above use the PCA-basis loss. For the protan participant, the sign of $\hat{\beta}_c$ (mechanism class) was confirmed under the PCA-basis loss only. The SRM-basis loss may yield a different mechanism class for this participant; sign stability under the SRM basis was not verified.

S17. Filter-evaluation session: design and comparator

Comparator. The deployed comparator was the macOS accessibility Color Filter (System Settings > Accessibility > Display, build 26.5.1), set at an intensity the participant self-tuned before scanning. It is the unmodified shipping product — an operating-system-level post-display transform. We compared against this deployed filter rather than a re-implemented retinal-simulation transform so that the comparator is the correction an end user actually encounters. The individualized filter, by contrast, was rendered directly in PsychoPy from the frozen per-subject pre-image.

Acquisition. Eight fMRI runs were acquired in ABBA-counterbalanced order, split evenly into four runs per filter. Scanner time set this count: a within-session two-condition design at six runs per condition would run roughly 90 minutes, beyond the 60–75-minute window over which participant attention and head motion stay acceptable. Four runs per condition (~ 60 minutes) sits at the recommended boundary.

Run-count adequacy. Four runs retain the Session-1 landmark, the separation between the healthy-control mean and the two CVD cases at hV4. Across all $\binom{6}{n}$ subsets of the six-run Session-1 data, hV4 LOCO adjacent accuracy held that separation

at every run count down to four (Figure S1). At $n = 4$ the healthy-control mean reached 0.45 against 0.46 at six runs and remained above the 0.25 chance level, while both CVD participants stayed below it (deutan 0.23, protan 0.14, single-case $d_{cc} < -2$ in both). A separate subsampling analysis of signal-quality and geometric-stability metrics returned the same ordering, with five runs preferred and four acceptable. For every neural index, the Session-1 HC reference runs were subsampled to four to match the per-condition run count, so condition and HC estimates rest on equal data.

Four runs sits below the count recommended for fine-grained RDM-based model comparison. The second session therefore targets movement toward or away from the HC reference rather than adjudication between model classes.

Single-case inference. Each condition’s neural index was compared to the HC distribution as a standardized single-case effect size (Cohen’s d against the HC mean and SD), reported in place of an inferential p because the grand-mean permutation null is biased for this design. Robustness was assessed in two ways: by leave-one-run separation (the contrast holding after any single run is dropped) and by re-extracting both conditions on the no-filter baseline voxel set.

Psychophysical battery. Hue-discrimination thresholds were estimated with JND staircases for eight hue pairs, and the eight-alternative forced-choice (8AFC) color-identification task comprised 64 trials. Both were compared to the unfiltered Session-1 baseline and to the HC distribution.

Color classification (LORO).. Table S14 reports LORO eight-way classification accuracy for every ROI and condition in both participants. Every cell stayed well above the 0.125 chance level, confirming that the displayed colors remained decodable and that neither filter degraded the second-session color signal.

Forward-tuning encoding index. Figure S2 shows the LOCO forward-tuning correlation ρ to the canonical hue map, a secondary encoding index reported as corroboration only (it lies outside the validated Session-1 decoding pipeline). In the deutan participant the individualized filter reaches the HC level at hV4 ($\rho = +0.18$ vs. HC $+0.21$) while the deployed filter drives it negative ($\rho = -0.39$); in the protan

participant the three conditions do not separate (all $\rho \approx -0.02$ at hV4).

S18. Statistical Analysis

Unless noted otherwise, the significance threshold was $\alpha = 0.05$. False Discovery Rate correction (Benjamini & Hochberg, 1995) was applied within two pre-specified families only: the RDM color-pair comparisons ($q = 0.05$; 28 pairs per participant) and the six parameter-identifiability checks (section 2.13). The exploratory per-hue vulnerability comparisons and the per-pair identification tests (including the blue-hue deficit) are reported one-tailed and uncorrected. Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (8)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Group-level effect sizes were quantified as Hedges' g (Hedges & Olkin, 1985). Single-case effect sizes are the case-control index d_{cc} defined in §S19. Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S19. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford & Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Effect sizes for the single-case interpolation comparisons are collected in table S15; the geometric-disparity effect sizes are listed alongside their t statistics in table S4 (case-control $d_{cc} = t\sqrt{8/7}$, $n = 7$). Single-case comparisons of within-subject LORO accuracy against the HC distribution reached $|d_{cc}|$ between 0.59 and 2.12 across the eight participant-by-ROI tests, with all $p \geq 0.095$. The largest deviation was the protan

participant at V3 ($d_{cc} = -2.12$), and at hV4 both participants gave $d_{cc} = -0.95$ ($p = 0.407$).

Cross-subject transfer of the forward encoding model was also compared between destinations. A control-trained model applied to a held-out control reached a mean accuracy of 0.526 over 28 subject-by-ROI cells, and the same model applied to a CVD participant reached 0.432 over 8 cells (Mann–Whitney $U = 163.5$, $p = 0.052$, rank-biserial $r = 0.46$). Both destinations lie far above the 0.125 chance level, and the lowest single cell reached 0.271. The comparison is reported in the SRM-aligned space, and sub-10 is excluded here as everywhere else.

Table S1

Procrustes disparity under the symmetric leave-one-subject-out reference, across the three preprocessing arms. Crawford–Howell one-tailed p .

Participant	ROI	Original	Motion regression	Shifted control
Protan	V1	0.045	0.022	0.031
Deutan	V2	0.116	0.300	0.044

Table S2

Eight-way classification accuracy under leave-one-run-out cross-validation, SRM-aligned amplitudes, chance = 0.125. Healthy-control mean $\pm$ SD ($n = 7$), with the two CVD participants listed separately. Bold: highest control mean in the ROI.

ROI	Decoder	HC (mean $\pm$ SD)	Deutan	Protan
V1	LDA	0.878 $\pm$ 0.050	1.000	0.854
	SVM	0.777 $\pm$ 0.062	0.979	0.771
	Forward encoding	0.542 $\pm$ 0.106	0.604	0.625
	Ridge	0.399 $\pm$ 0.054	0.354	0.250
	Kernel ridge	0.393 $\pm$ 0.064	0.271	0.292
	MLP	0.128 $\pm$ 0.008	0.125	0.125
V2	LDA	0.830 $\pm$ 0.047	0.917	0.854
	SVM	0.759 $\pm$ 0.136	0.875	0.833
	Forward encoding	0.545 $\pm$ 0.077	0.458	0.438
	Ridge	0.372 $\pm$ 0.097	0.458	0.250
	Kernel ridge	0.330 $\pm$ 0.066	0.313	0.250
	MLP	0.149 $\pm$ 0.031	0.125	0.125
V3	LDA	0.726 $\pm$ 0.064	0.750	0.771
	SVM	0.631 $\pm$ 0.144	0.813	0.667
	Forward encoding	0.449 $\pm$ 0.065	0.375	0.313
	Ridge	0.173 $\pm$ 0.068	0.292	0.229
	Kernel ridge	0.158 $\pm$ 0.052	0.208	0.333
	MLP	0.122 $\pm$ 0.008	0.125	0.125
hV4	LDA	0.658 $\pm$ 0.092	0.813	0.729
	SVM	0.598 $\pm$ 0.085	0.813	0.667
	Forward encoding	0.423 $\pm$ 0.072	0.354	0.354
	Ridge	0.319 $\pm$ 0.114	0.146	0.250
	Kernel ridge	0.277 $\pm$ 0.093	0.042	0.188
	MLP	0.125 $\pm$ 0.000	0.125	0.125

Table S3

Adjacent accuracy under leave-one-color-out cross-validation, SRM-aligned amplitudes. Healthy-control mean $\pm$ SD ($n = 7$), with the two CVD participants listed separately. Chance depends on the output space of the decoder. Forward encoding, ridge and kernel ridge return a continuous hue, so a uniform prediction lands within the 45° tolerance on 91 of 360 draws and chance is 0.25. The LDA, SVM and MLP classifiers return one of the eight stimulus hues, for which the tolerance admits 3 of 8 and chance is 0.375. The constant-output MLP sits at exactly 0.375 at V1 and V2, which confirms the classifier value. Bold: the largest margin above the applicable chance level. Kernel ridge and the MLP are constant-output rows, noted in the text.

ROI	Decoder	HC (mean $\pm$ SD)	Deutan	Protan
V1	Forward encoding	0.360 $\pm$ 0.098	0.313	0.250
	LDA	0.268 $\pm$ 0.074	0.375	0.167
	SVM	0.226 $\pm$ 0.078	0.375	0.229
	Ridge	0.039 $\pm$ 0.053	0.146	0.000
	MLP	0.375 $\pm$ 0.012	0.375	0.375
	Kernel ridge	0.000 $\pm$ 0.000	0.000	0.000
V2	Forward encoding	0.283 $\pm$ 0.101	0.333	0.063
	LDA	0.327 $\pm$ 0.088	0.313	0.417
	SVM	0.318 $\pm$ 0.108	0.250	0.396
	Ridge	0.054 $\pm$ 0.083	0.021	0.000
	MLP	0.375 $\pm$ 0.000	0.396	0.375
	Kernel ridge	0.000 $\pm$ 0.000	0.000	0.000
V3	Forward encoding	0.220 $\pm$ 0.100	0.354	0.125
	LDA	0.220 $\pm$ 0.081	0.271	0.333
	SVM	0.208 $\pm$ 0.103	0.250	0.229
	Ridge	0.000 $\pm$ 0.000	0.000	0.000
	MLP	0.250 $\pm$ 0.000	0.250	0.250
	Kernel ridge	0.000 $\pm$ 0.000	0.000	0.000
hV4	Forward encoding	0.470 $\pm$ 0.130	0.271	0.104
	LDA	0.280 $\pm$ 0.144	0.146	0.229

Table S4

*Procrustes disparity, Crawford–Howell one-tailed single-case t ($df = 6$) against the HC reference, in the primary common HC-trained SRM space and under the symmetric LOSO control, with the common-space case-control effect size $d_{cc} = t\sqrt{8/7}$ ($n = 7$).
Bold: $p < 0.05$.*

Subject	ROI	Common space			Symmetric LOSO	
		t	p	d_{cc}	t	p
Deutan	V1	1.10	0.157	1.18	0.48	0.323
	V2	2.11	0.040	2.26	1.33	0.116
	V3	1.92	0.052	2.05	1.17	0.143
	hV4	0.23	0.411	0.25	0.07	0.474
Protan	V1	3.48	0.007	3.72	2.02	0.045
	V2	0.99	0.181	1.06	0.77	0.234
	V3	0.09	0.466	0.10	0.06	0.479
	hV4	1.13	0.150	1.21	0.80	0.228

Table S5

Color-label permutation on the healthy controls, used as a positive control. Detection counts and mean permutation z under a projection re-estimated at each permutation and under a frozen projection. Lower z indicates a better identity correspondence than the shuffled null.

ROI	n	Re-estimated		Frozen	
		detected	mean z	detected	mean z
V1	7	0	−0.04	2	−1.58
V2	7	0	−0.04	3	−1.00
V3	7	0	−0.78	5	−2.39
hV4	6	0	+0.06	2	−1.07

Table S6

Color-label permutation under the frozen projection, one cell per participant and ROI.

Entries give p_{perm} with the Benjamini–Hochberg q in parentheses, corrected across all 35 cells. Bold: $q < 0.05$. HC 7 lacks sufficient hV4 coverage.

Participant	V1	V2	V3	hV4
HC 1	0.088 (0.162)	0.020 (0.064)	0.062 (0.120)	0.393 (0.443)
HC 2	0.001 (0.012)	0.016 (0.056)	0.004 (0.023)	0.136 (0.207)
HC 3	0.109 (0.181)	0.520 (0.552)	0.255 (0.329)	0.272 (0.329)
HC 4	0.152 (0.221)	0.014 (0.054)	0.001 (0.012)	0.041 (0.090)
HC 5	0.057 (0.117)	0.773 (0.773)	0.034 (0.079)	0.031 (0.079)
HC 6	0.407 (0.445)	0.290 (0.338)	0.009 (0.045)	0.220 (0.296)
HC 7	0.034 (0.079)	0.159 (0.222)	0.004 (0.023)	—
Deutan	0.105 (0.181)	0.002 (0.018)	0.024 (0.070)	0.273 (0.329)
Protan	0.758 (0.773)	0.013 (0.054)	0.001 (0.012)	0.129 (0.205)

Table S7

Convergent validity of SRM disparity with two alignment-independent distances,

Spearman r over the nine analysed participants. Crossnobis distance uses native voxel space with no alignment step. PCA and PCA-CCA replace the shared response model with a pairwise alignment at matched dimensionality. Bold: $p < 0.05$.

Distance	V1	V2	V3	hV4	Pooled
Crossnobis (voxel space)	0.833	0.733	0.550	0.300	0.632
PCA	0.633	0.850	0.333	0.533	0.780
PCA-CCA	0.483	0.433	0.100	−0.067	0.544

Table S8

Variance explained by the shared response model under the symmetric leave-one-subject-out projection, in which controls and CVD participants are projected identically. Hedges' g is signed control minus CVD, so negative values indicate better reconstruction of the CVD data.

ROI	k	Controls	CVD	g
V1	4	0.352	0.407	-0.40
V2	4	0.331	0.416	-1.26
V3	3	0.250	0.314	-0.66
hV4	3	0.225	0.253	-0.39

Table S9

Distortion-model fits for both CVD participants, all scored on the same held-out composite test-loss ($\bar{L}_{\text{test}}$, lower is better; §2.12; the neural RDM atom in each adopted fit is V2 for the deutan and V1 for the protan participant). The 2-component model is shown at its adopted fit. The retinal-plus-cortical (R+C) fits are reported for comparison: the deutan R+C was rejected at the boundary-saturation gate (§3.4), and the protan R+C generalized worse than the 2-component fit.

Participant	Model	Fitted parameters	Held-out fit
Deutan	2-component	$\beta_s = 6^\circ$, $\beta_c = -42^\circ$	$\bar{L}_{\text{test}} = -2.36$
Protan	2-component	$\beta_s = 2^\circ$, $\beta_c = +24^\circ$	$\bar{L}_{\text{test}} = -1.54$
Deutan	R+C	$g \rightarrow 3.0$ (100% saturated)	rejected (Gate 2)
Protan	R+C	$g = 2.95$ (41% saturated)	$\bar{L}_{\text{test}} = -0.86$

Candidate	bias (β_s, β_c)	f_{10°	Verdict
S08-robust ($6^\circ, -42^\circ$)	($+16^\circ, -4.7^\circ$)	0.26	FAIL
S09-primary ($2^\circ, +24^\circ$)	($+11^\circ, -27^\circ$)	0.14	FAIL

Table S10

Test 1 results. Recovery reached $f_{10^\circ} = 0.26$ and 0.14 against a pre-specified criterion of 0.5 . The non-dominant axes ($|\hat{\beta}_s| \leq 6^\circ$ for both participants) fall below the effective parameter uncertainty established by Test 2a and are not recoverable. For S08-robust, the dominant axis ($|\hat{\beta}_c| = 42^\circ$, bias = 4.7°) shows lower bias than the non-dominant axis (16°), consistent with partial magnitude recoverability where the axis magnitude exceeds the noise floor.

Candidate	$ \hat{\beta}_s _{\text{med}}$ (IQR)	$ \hat{\beta}_c _{\text{med}}$ (IQR)	β_c p95	$f_{10^\circ}^{\text{origin}}$
S08-robust	22° (40)	26° (10.5)	44°	0.00
S09-primary	16° (17.5)	24° (9)	48°	0.00

Table S11

Test 2a results. $f_{10^\circ}^{\text{origin}} = 0/140$ for both candidates: no sample was recovered within 10° of the origin under a null ground truth. Effective parameter uncertainty (PCA-basis): $\sim 20^\circ$ (β_s) and $\sim 25^\circ$ (β_c).

Candidate	rank _{dist}	Verdict
S08-robust	0.875	FAIL
S09-primary	0.875	FAIL

Table S12

Test 2b results. Neither candidate exceeds all HC pseudo-CVD distances.

Candidate	Real loss	Perm 5% cut	p_{perm}
S08-robust	-2.892	-3.136	0.167
S09-primary	-1.681	-3.053	0.471

Table S13

Test 2c results. Neither candidate reaches significance against the label-permutation null.

Table S14

Second-session color classification (LORO eight-way classification accuracy). *Chance = 0.125. NF, no-filter (Session-1 baseline); Dep, deployed accessibility filter; Ind, individualized filter. HC, healthy-control mean ($n = 7$).*

ROI	HC	Deutan (sub-08)			Protan (sub-09)		
		NF	Dep	Ind	NF	Dep	Ind
V1	0.71	0.79	0.84	0.72	0.79	0.91	0.66
V2	0.71	0.79	0.69	0.62	0.83	0.75	0.72
V3	0.77	0.67	0.78	0.69	0.81	1.00	0.69
hV4	0.75	0.73	0.88	0.50	0.71	0.84	0.69

Table S15

Effect sizes for the single-case hue-interpolation comparisons at hV4 (main text, §3.2). Crawford–Howell case-control effect size $d_{cc} = t\sqrt{(n+1)/n}$ against the $n = 6$ HC distribution. Per-hue tests are one-tailed, uncorrected, and exploratory; both CVD participants share identical per-hue statistics where adjacent accuracy is zero. No individual hue reaches significance.

Comparison (hV4)	t	p	d_{cc}
LOCO adjacent accuracy, deutan	−1.84	0.063	−1.99
LOCO adjacent accuracy, protan	−2.91	0.017	−3.14
<i>Per-hue adjacent accuracy (one-tailed, exploratory; both participants)</i>			
Blue	−1.73	0.072	−1.87
Purple	−0.90	0.205	−0.97
Magenta	−1.32	0.122	−1.43

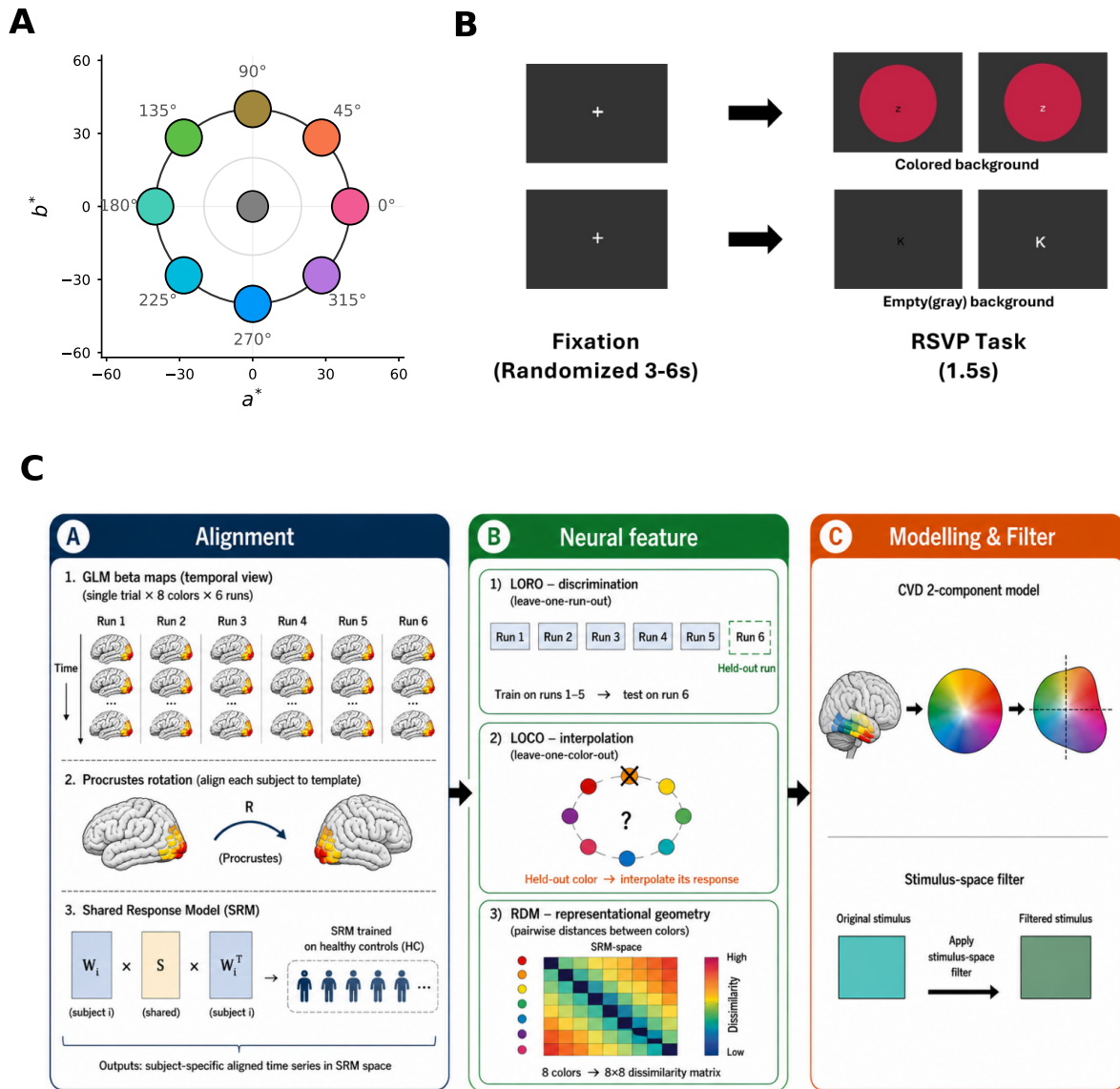


Figure 1. Stimuli, task, and analysis pipeline. (A) Stimulus set: 8 CIE $L^*a^*b^*$ hues at matched lightness, 45° angular spacing ($L^* = 75$, chroma = 40; red–magenta), shown in the a^*b^* plane. (B) Task: rapid serial visual presentation (RSVP). A fixation period (randomized 3–6s) is followed by a 1.5s uniform colored disc; 5–9 trials per color per run (Neurodesign-optimized allocation, identical across participants), six runs. (C) Analysis pipeline. Stage A (Alignment): two-stage GLM per-color amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Stage B (Neural feature): LORO classification, LOCO interpolation, and the SRM-space representational dissimilarity matrix (RDM). Stage C (Modeling & filter): a 2-component cortical model with S-cone-axis (β_s) and confusion-axis (β_c) components; its pre-image yields the per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07).

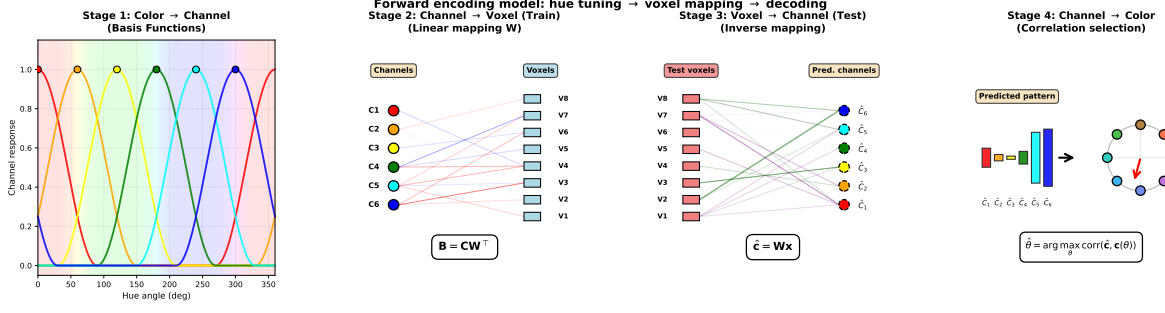


Figure 2. Forward encoding model schematic. Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): the channel-to-voxel weights W are fit by ordinary least squares for hue decoding, or by GCV-regularized ridge regression for voxel-response prediction (encoding). Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as $\hat{c} = Wx$. Stage 4: the decoded hue is recovered as the angle, among all 360 integer hues, whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same model serves both decoding schemes (Section 2.8): cross-run classification (LORO) and cross-color interpolation (LOCO). Their basis and correlation-based selection are identical, and only the held-out test set differs. Decoding uses the OLS pseudoinverse weights.

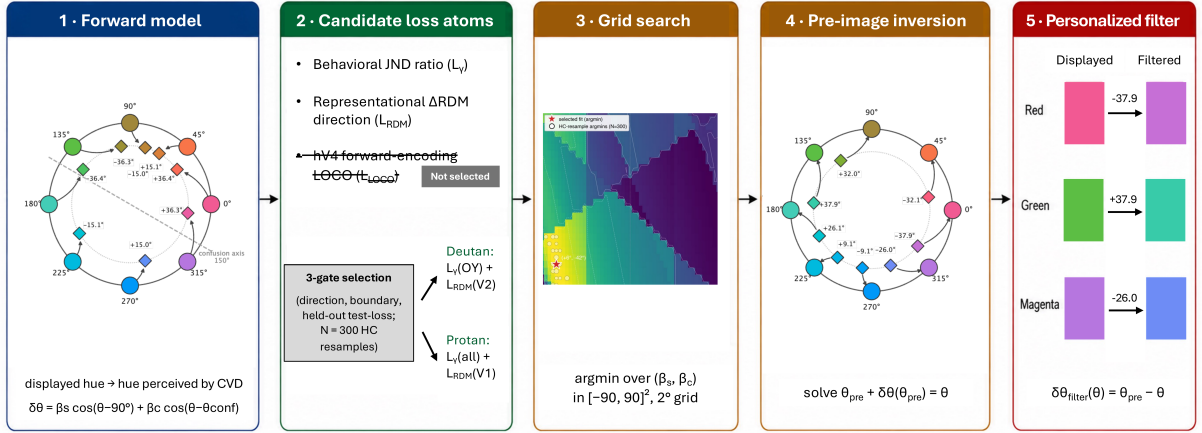


Figure 3. Fitting a per-subject cortical distortion and inverting it into a stimulus-space filter. (1) A two-parameter forward model expresses the per-hue distortion as $\delta\theta(\theta) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}})$: an S-cone-axis component (β_s) plus a family-specific confusion-axis rotation (β_c ; $\theta_{\text{conf}} = 16^\circ$ protan, 150° deutan). Outer ring, displayed hue; inner ring, hue perceived under the fitted distortion; labels give the signed displacement. (2) Three candidate loss atoms quantify CVD–HC differences. A three-gate procedure (direction, boundary saturation, held-out test-loss over $N = 300$ HC resamples; §2.12) selected a different combination for each participant (deutan, $\gamma_{\text{OY}} + L_{\text{RDM}}^{(\text{V2})}$; protan, $\gamma_{\text{all}} + L_{\text{RDM}}^{(\text{V1})}$); the LOCO atom entered neither selected combination. (3) A grid search over $(\beta_s, \beta_c) \in [-90^\circ, 90^\circ]^2$ at 2° resolution minimizes the z -scored composite to the selected fit $\hat{\beta}$. The surface shown is the deutan participant’s; both landscapes and their resample uncertainty are in Figure 6. (4) The fitted forward map is numerically inverted: for each displayed hue θ , the corrected stimulus θ_{pre} solves $\theta_{\text{pre}} + \delta\theta(\theta_{\text{pre}}) = \theta$. (5) The per-subject filter is $\delta\theta_{\text{filter}}(\theta) = \theta_{\text{pre}} - \theta$; three of the eight hues are shown for the deutan participant, and the full eight-hue filter for both participants is in Figure 7. Hue wheels and swatches are computed from each participant’s fitted parameters; the schematic reports no new statistics.

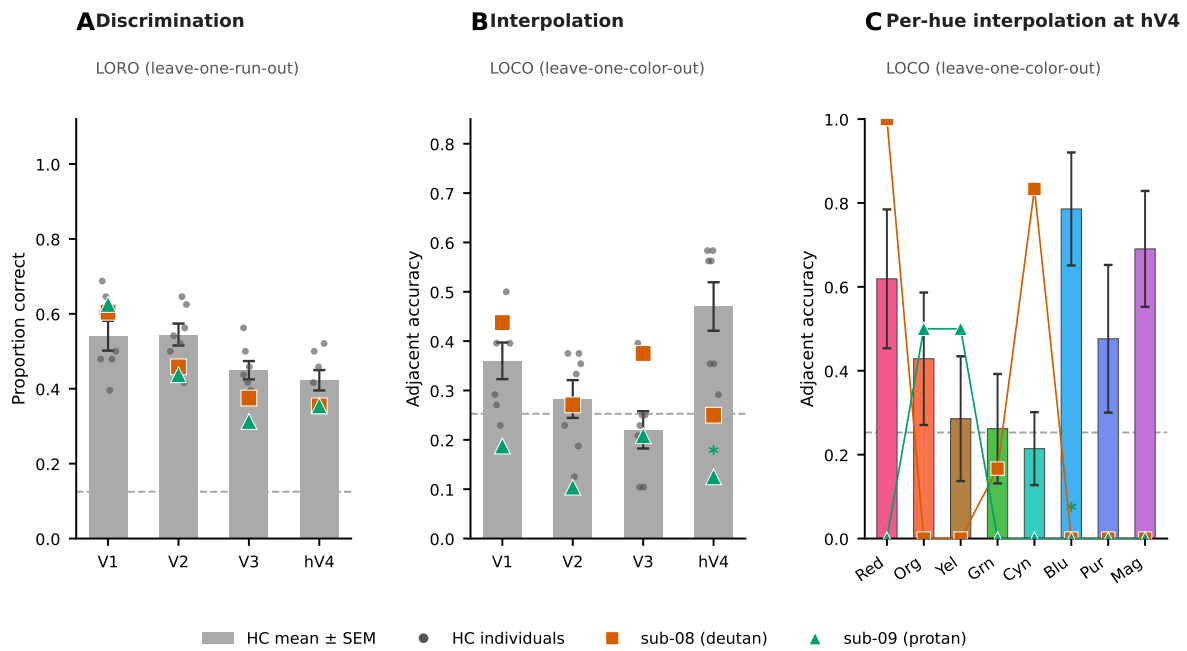


Figure 4. Color classification and hue interpolation across V1–hV4 in the two CVD participants and healthy controls. (A) LORO eight-way classification accuracy (forward encoding model, SRM-aligned), V1–hV4. HC mean $\pm$ SEM ($n = 7$; $n = 6$ at hV4, sub-07 excluded for low voxel count; gray); the deutran participant (orange square), the protan participant (teal triangle). Chance = $1/8$. **(B)** LOCO adjacent accuracy (prediction within 45° of the target hue; chance = 0.25). *: the protan participant below HC at hV4 (Crawford & Howell $t = -2.91$, $p = 0.017$; exploratory single-case finding); the deutran participant $t = -1.84$, $p = 0.063$, n.s. **(C)** Per-hue adjacent accuracy at hV4, HC bars $\pm$ SEM ($n = 6$). Per-hue single-case tests (Crawford & Howell, one-tailed, uncorrected and exploratory) reached significance at no individual hue; the deficit concentrated on the S-cone intermediate hues blue, purple, and magenta, where both participants showed zero adjacent accuracy. Joint pattern in (A) and (B–C) defines the filter precondition and the functional vulnerability profile, respectively.

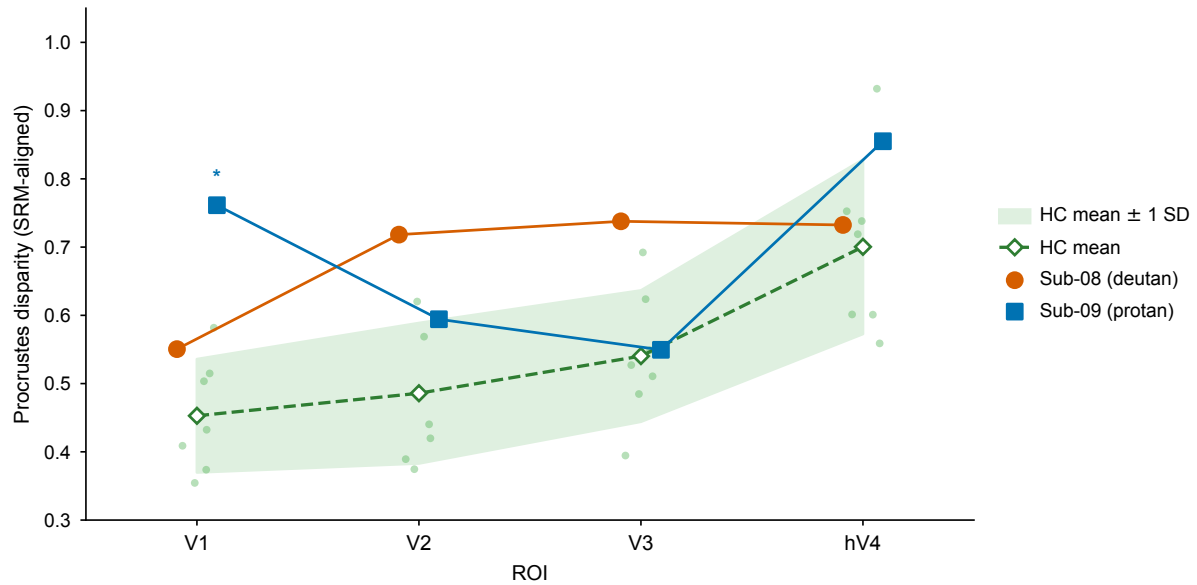


Figure 5. Procrustes disparity to the healthy-control reference. Disparity per participant per ROI, which indexes the distance of the whole geometry from that reference. HC band, group mean ± 1 SD ($n = 7$); dots, individual HC leave-one-out values. Single-case comparisons used the Crawford–Howell modified t against the HC leave-one-out distribution. The asterisk marks the one comparison that reaches $p < 0.05$ under the symmetric LOSO reference, which is the inferential estimate (§2.9); test statistics under both references are given in table S4. Exploratory single-case.

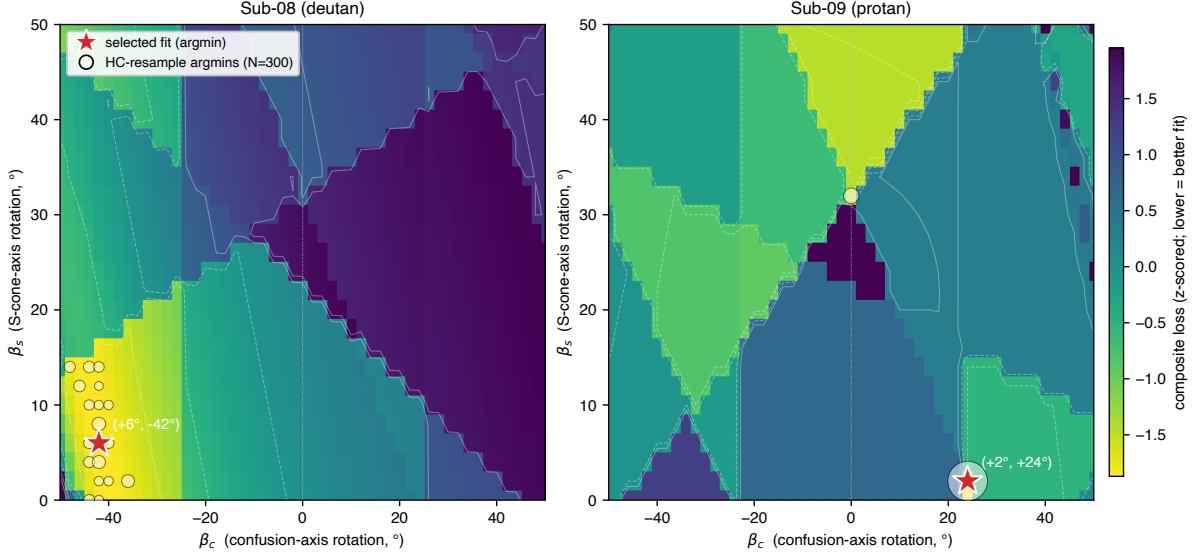


Figure 6. Per-subject loss landscape and parameter uncertainty. z -scored composite loss over the (β_s, β_c) grid for each CVD participant’s selected loss combination (the deutan participant: $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$; the protan participant: $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$), reconstructed on the full 7-HC pool. Lower (yellow) indicates better fit. Red star: selected fit (argmin) — the deutan participant $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$; the protan participant $(2^\circ, +24^\circ)$. White points: per-resample argmins over $N = 300$ HC 5-train/2-test resamples, showing parameter uncertainty. The selected fits are descriptive low-dimensional embeddings of each participant’s distortion, not physiological point estimates (this section).

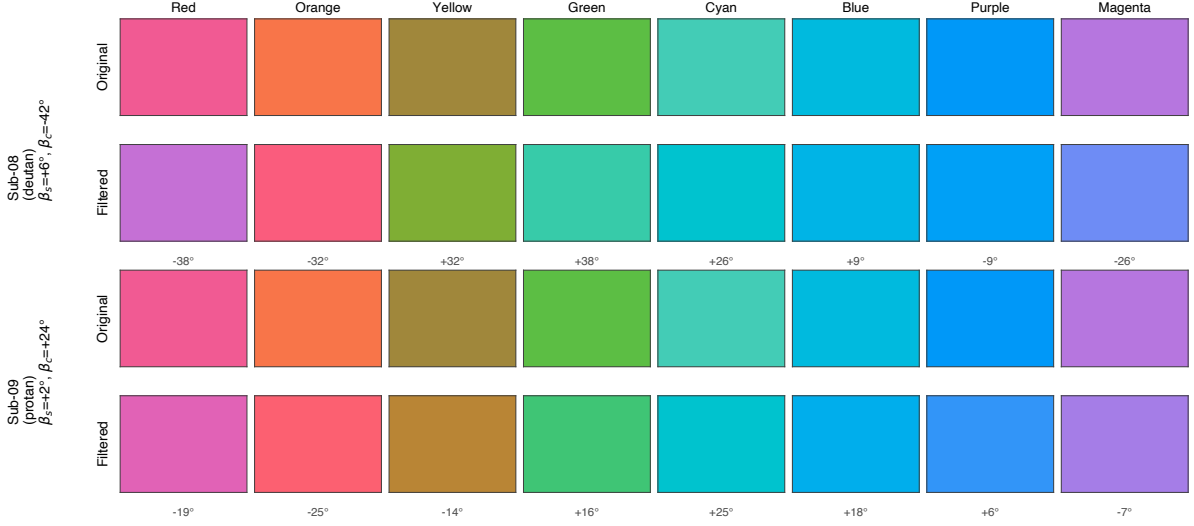


Figure 7. Per-subject stimulus-space correction filter. For each CVD participant, the displayed stimulus (*Original*) and the corrected stimulus (*Filtered*) are shown for the eight scanner hues, which run across the columns. Top block: the deutan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$. Bottom block: the protan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$. The filtered stimulus is the exact numerical pre-image of the fitted 2-component transform; the per-hue correction $\delta\theta_{\text{filter}} = \theta_{\text{pre}} - \theta$ is annotated beneath each *Filtered* swatch (8/8 pre-images exact, residual $< 0.001^\circ$ for both subjects). Rendering coordinate space: STIM_LAB CIELab (project convention). Neural validation (hue-interpolation adjacent accuracy and representational geometry) and psychophysical validation (JND thresholds and 8AFC color identification) in second sessions in both participants are reported in fig. 8; replication in more individuals remains for future work.

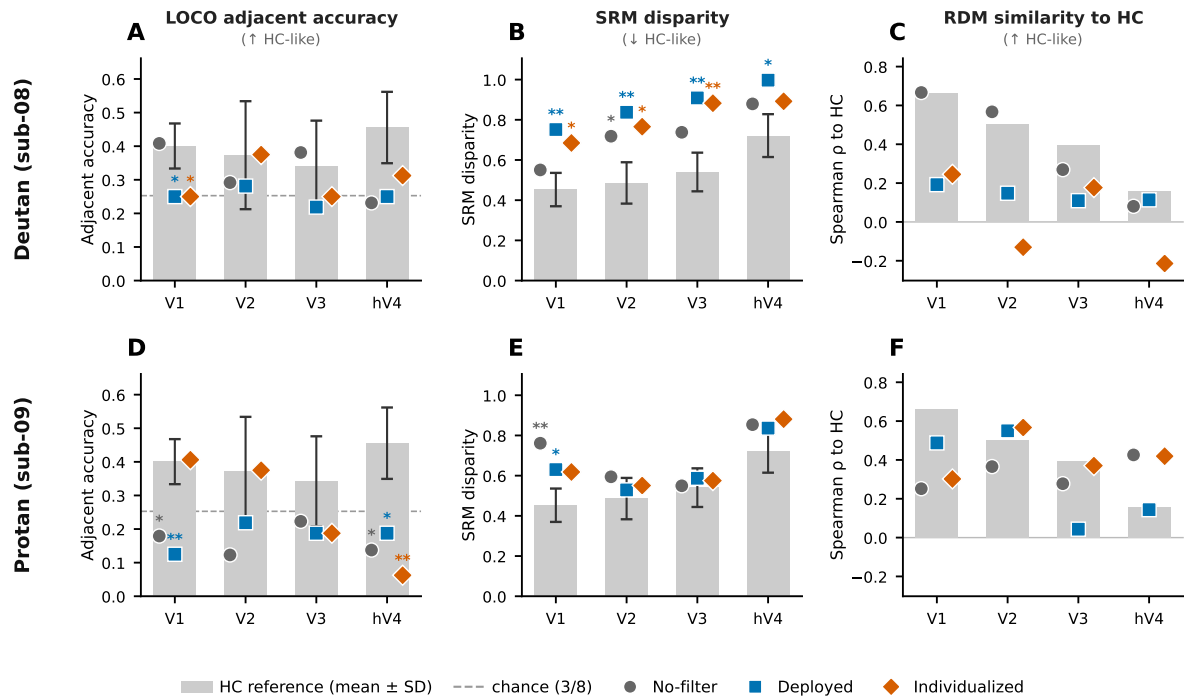


Figure 8. Hue interpolation and representational geometry under each filter in the second session. Rows: deutan (sub-08, A–C) and protan (sub-09, D–F). Columns: interpolation and representational geometry across V1–hV4. Gray bars, HC reference (mean $\pm$ SD, $n = 7$). Markers: no-filter (gray circle, Session-1 baseline), deployed accessibility filter (blue square), individualized filter (orange diamond); dashed line, chance. **(A, D)** LOCO adjacent accuracy (± 1 hue step; interpolation; $\uparrow$ HC-like; chance = 0.25; hV4 is the validated ROI). **(B, E)** SRM disparity into the HC shared space ($\downarrow$ HC-like). **(C, F)** Shared-space RDM similarity to HC ($\uparrow$ HC-like). Filter effects on hV4 interpolation differed between the two participants (individualized filter: deutan improved, protan reduced), and neither filter reproduced an HC-like color geometry. Single case per subtype, descriptive. Stars, Crawford–Howell single-case deviation from the HC distribution (* $p < .05$, ** $p < .01$, *** $p < .001$); the forward-tuning encoding index is in Fig. S2.

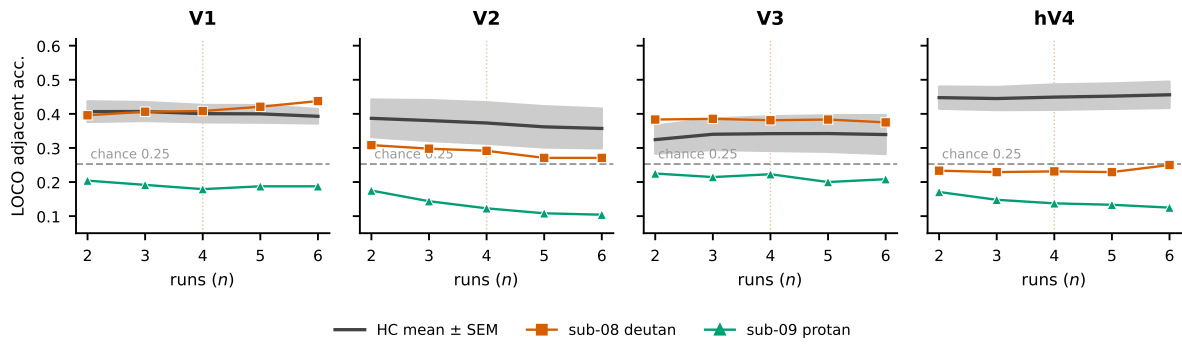


Figure S1. Hue-interpolation adjacent accuracy as a function of the number of runs. LOCO adjacent accuracy (canonical FE-6 basis, OLS pseudoinverse decoder) as a function of the number of runs n , computed over all $\binom{6}{n}$ run subsets of the Session-1 data, per ROI. Black line, healthy-control mean $\pm$ SEM (shaded band); dashed line, the 0.25 chance level; markers, the two CVD single cases (sub-08 deutan, sub-09 protan). At hV4 the healthy-control mean stays above chance and the CVD deficit persists at every run count, including the deployed $n = 4$ (orange guide); the metric does not depend on having the full six runs.

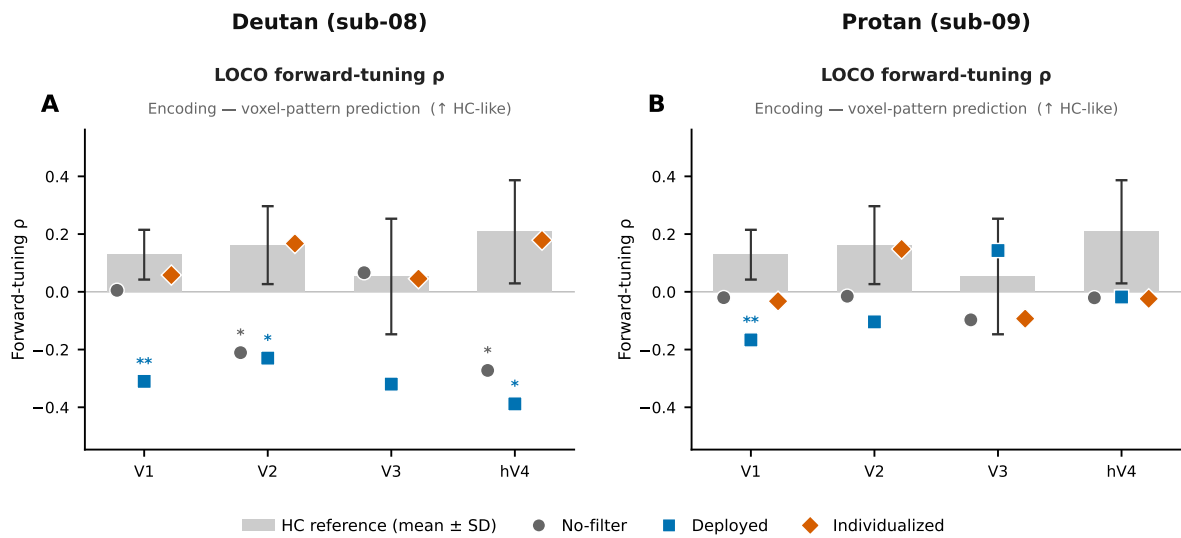


Figure S2. Forward-tuning encoding index (corroboration only). LOCO forward-tuning correlation ρ to the canonical hue map across V1–hV4, for the deutan (sub-08, left) and protan (sub-09, right) participants ($\uparrow$ HC-like). Gray bars, HC reference (mean $\pm$ SD, $n = 7$); markers, no-filter (gray circle), deployed accessibility filter (blue square), individualized filter (orange diamond). Stars, Crawford–Howell single-case deviation from the HC distribution ($*p < .05$, $**p < .01$).